A Web-Based In-Kind Inventory Management System for Local Community Donation Centres

219969 & 219723

A System Documentation Submitted to the Strathmore Institute of Management and Technology in partial fulfilment of the requirements for the award of a Diploma of Business and Information Technology

Diploma in Business Information and Technology

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Declaration and Approval

Declaration

We declare that this work has not been previously submitted and approved for the award of a Diploma in Business Information and Technology by this or any other University. To the best of our knowledge and belief, the documentation contains no material previously published or written by another person except where due reference is made in the documentation itself.

Student's signature:

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Approval

The Information System Project documentation of 219969 and 219723 was reviewed and approved (for examination) by:

Supervisor's signature:

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Abstract

In developing countries like Kenya, many local community donation centres face significant logistical and accountability challenges when managing in-kind donations as a result of reliance on manual, paper-based systems that lead to delays, errors, resource wastage, and distrust from donors, hindering their ability to provide aid. Review of existing solutions like HELM, PlanStreet, and WASP provided systems that solve the problem but had budget and technological constraints for smaller non-profit organisations.

To address these gaps, this project was aimed to develop a centralised, cost-effective, and easy-to-navigate system that ensured transparency and traceability throughout an item's lifecycle. Using the Agile-Scrum methodology with HTML, JavaScript, CSS, PHP, and MySQL stack to develop the fully functional system that consisted of donor, staff, and admin modules, each having the ability to make and tack donations, manage inventory with QR codes, and manage users, respectively.

Despite achieving full system functionality, a significant limitation was the failure to apply automated testing to validate the non functional requirements: scalability, capacity, and reliability as a result of technical skill gaps which lessened the quality of the final system. Nonetheless, the overall project delivers a meaningful contribution by providing accessible, budget-friendly, and sustainable digital solution that improves the efficiency and accuracy when managing in-kind donation inventory in a local community donation centre.

Table of Contents

Declaration and Approval ii

Abstract iii

List of Figures ix

List of Tables x

List of Abbreviations xi

Chapter 1: Introduction 1

1.1 Background 1

1.2 Problem Statement 2

1.3 Aim/General Objective 3

1.4 Specific Objectives 3

1.5 Justification 4

1.6 Scope 4

1.7 Limitations and Delimitations 5

Chapter 2: Literature Review 6

2.1 Introduction 6

2.2 Obstacles Encountered by Local Community Donation Centres in Managing Non-Monetary Donations 6

2.2.1 Reliance on Inadequate, Manual Systems 6

2.2.2 Strategic Requirements and Organisational Deficiency 7

2.2.3 Consequences of Insufficient Systems 7

2.3 Challenges of Current Donation Inventory Management Solutions 7

2.3.1 Reliance on Manual, Error-Prone Processes 7

2.3.2 Risk of Perishable Item Wastage 8

2.3.3 Lack of Integration Between Systems 8

2.4 Review of Existing Systems 9

2.4.1 Humanitarian Enterprise Logistics Manager (HELM) 9
2.4.2 PlanStreet 12
2.4.3 WASP Inventory Cloud 13
2.5 Conceptual Framework 15
Chapter 3: Methodology 17
3.1 Introduction 17
3.2 Research Approach 17
3.3 Methodology 17
3.3.1 Foundational Phase: Requirements Gathering and Product Backlog Creation 18
3.3.2 Planning and Design Phase: Object-Oriented Design and Architectural Planning 19
3.3.3 Implementation Phase: Development Environment and Tools Utilised 19
3.3.4 Testing Phase: Testing the System's Quality and Validation 20
3.4 Deliverables 20
3.4.1 Documentation 20
3.4.2 System Design Artifacts 21
3.4.3 Core Application Modules 21
Chapter 4: System Analysis and Design 22
4.1 Introduction 22
4.2 Requirement Analysis 22
4.2.1 Functional Requirements 22
4.2.2 Non-Functional Requirements 23
4.3 Diagrams 26
4.3.1 Use Case Diagram 26
4.3.2 Class Diagram 27
4.3.3 Sequence Diagrams 28

4.3.4 Entity Relationship Diagram 30
4.3.5 Database Schema 31
4.3.6 Wireframes 32
Chapter 5: Development and Testing 38
5.1 Introduction 38
5.2 Setup Description 38
5.2.1 Solution Architecture 38
5.2.2 Test Environment 39
5.2.3 Operational Workflow 40
5.2.4 Stakeholder Interaction 41
5.2.5 Deviations from Proposed Implementations 42
5.2.6 Challenges 43
5.3 Testing 44
5.3.1 Test Cases 44
5.3.2 Test Case Results 52
Chapter Six: Conclusions and Recommendations 56
6.1 Conclusions 56
6.2 Recommendations 58
6.2.1 System Testing 58
6.2.2 Further Inventory Categorisation and Enhanced Distribution 58
6.2.3 Live Chat Communication 59
6.2.4 Incorporations of Multiple Centres into One System 59
References 60
Appendices 63
Appendix i: PHP QR Code 63

Error Message Appendices....63
Appendix 1: Staff/admin login error....63
Appendix 2: More than max quantity error....63
Non-Functional Requirements Appendices....64
Appendix 3: Portability....64
Test Case Result Appendices....64
Appendix A: Successful Donor Login....64
Appendix B: Registration Reflected in Database....65
Appendix C: Successful Email Verification Sent....65
Appendix D: Donation Form....65
Appendix E: Donation Tracking....65
Appendix EA: Donor Feedback....66
Appendix EB: Donor Report Generation....66
Appendix F: Successful Staff Login....67
Appendix G: Rejected Staff Login....67
Appendix H: Adding a Critical Item....67
Appendix I: Deleting a Critical Item....67
Appendix J: Editing Donation Status....68
Appendix K: When Donation Status is 'In Inventory'....68
Appendix KA: Past Date as an Expiry Date....69
Appendix KB: Negative Quantity Error....69
Appendix KC: Cannot Change Donation Status Once Distributed....70
Appendix L: Staff Report Generation....70
Appendix M: Staff Feedback....70
Appendix N: Successful Admin Login....71

Appendix NA: Rejected Admin User 71
Appendix O: User Approvals 71
Appendix P: Role Changing 71
Appendix Q: Deleting a Donor User 72
Appendix R: Admin Report Generation 72
Appendix S: Admin Feedback 72
Appendix T: System QR Code Generation 73
Appendix U: System QR Code Scanning 73

List of Figures

Figure 2.1 Application Architecture of the Humanitarian Enterprise Logistics Manager (HELM), (HELM - Humanitarian Software, 2025) 10
Figure 2.2 PlanStreet's Inventory Management System, (Inventory Management Software, 2025) 12
Figure 2.3 New Inventory Item on Inventory Cloud by WASP, (Non Profit Inventory Management Made Simple, 2025) 14
Figure 2.4 Conceptual Diagram 16
Figure 3.1 Scrum Process Mapping for Project Development 18
Figure 4.1 Use Case Diagram 26
Figure 4.2 Class Diagram 27
Figure 4.3 Staff Sequence Diagram 28
Figure 4.4 Donor Sequence Diagram 29
Figure 4.5 Entity Relationship Diagram 30
Figure 4.6 Database Schema 31
Figure 4.7 Login Page Wireframe 32
Figure 4.8 Donor Registration Page Wireframe 33
Figure 4.9 Staff/Admin Registration Page Wireframe 34
Figure 4.10 Donor Page Wireframe 35
Figure 4.11 Staff and Volunteer Page Wireframe 36
Figure 4.12 Admin Page Wireframe 37

List of Tables

Table 5.1 Test Environment Table 39
Table 5.2 Donor Module Test Cases 44
Table 5.3 Staff Module Test Cases 46
Table 5.4 Admin Module Test Cases 49
Table 5.5 Systems Test Cases 51
Table 5.6 Donor Module Test Case Results 52
Table 5.7 Staff Module Test Case Results 53
Table 5.8 Admin Module Test Case Results 54
Table 5.9 System Test Case Results 55

List of Abbreviations

API – Application Programming Interface

CSS – Cascading Style Sheets

ERD – Entity-Relationship Diagram

HELM – Humanitarian Enterprise Logistics Manager

HTML – HyperText Markup Language

OOAD – Object-Oriented Analysis and Design

PHP – Hypertext Preprocessor

SaaS – Software as a Service

SQL – Structured Query Language

WMS – Warehouse Management System

QR – Quick Response

RESTful – Representational State Transfer

SMS – Short Message Service

ERP – Enterprise Resource Planning

BI – Business Intelligence

UI – User Interface

Chapter 1: Introduction

1.1 Background

Local community donation centres are the mediators between donors and vulnerable populations. Especially in developing nations like Kenya, where they play a crucial role in distributing essential resources. While most of the donations are monetary, there is still a large portion of non-monetary donations, also known as in-kind donations, which include goods like food, medicine, textiles, hygiene products, and much more (Islam, 2013).

One of the biggest challenges faced by non-profit organisations is insufficient inventory management, with only 14.6% of Kenyan non-governmental organisations agreeing that their operations are well-documented and that performance gaps are evaluated regularly (Kaberi, 2013). This issue stems from the reliance of outdated, unsecure, and error-prone methods including spreadsheets and paper-based operations, which cannot adapt to growing operations (Shapiro, Oystrick, Liu, & Vadlamudi, 2024), and hinder an organisation's ability to efficiently and effectively distribute goods because of significant operational issues such as delays, errors, and wastage of resources (Ndei & Mutuku, 2021).

This problem affects multiple parties involved: Staff and volunteers are burdened with manual, time-consuming tasks that divert their attention and time from more important work. Donors may become hesitant to donate again if there is a lack of transparency and exclusion from the process. The organisation's reputation is at risk from its ineffective operations that waste resources and affect its accountability. Finally, the recipients of the donations suffer the most, because the inefficiencies result in understocking of essential items that leave centres unable to meet the demands of those in need (Islam, 2013).

To solve this problem, some researchers helped a non-profit organisation adopt a database solution from an Excel-based data management system. This solution provided a more structured and robust management of inventory that allowed the organisation to collect, manage, and analyse data efficiently. Such a solution included data validation, data integrity, and security, allowing for more complex data analysis that enabled data-based decision making in the organisation (Shapiro, Oystrick, Liu, & Vadlamudi, 2024). Another solution proposed by Florence Muthoni Ndei and Morrisson Mutuku is an e-procurement system to manage donations electronically by digitally logging and categorising donations upon arrival, tracking the entire lifecycle of each of the donated

items, and automatically generating reports on inventory levels and distribution to improve the organisation's performance and accountability to donors (Ndei & Mutuku, 2021). Finally, Muniandy and Arbaiy's review of existing solutions, including: Inventory Storage Management Application, Simple Inventory Control, and Pocket Inventory Management revealed that the solution offered some features, such as a login and registration module, inventory management, internet requirements, and a report generator; however none of the solutions incorporated all the comprehensive features (Muniandy & Arbaiy, 2022).

These solutions don't provide a fully integrated system that comprises all the necessary features and functionalities required in an ideal inventory system. Moreover, these solutions often cater to a single product category and lack the versatility to manage a diverse range of donated goods. Finally, there is the complete lack of donor engagement in these solutions. which prevents transparency, trust, and bonds with the donors. To address these gaps, a web-based application which integrates numerous features of inventory management, accommodates all types of non-monetary donations, and involves donors in how their contributions are being used needs to be developed.

1.2 Problem Statement

Many local community donation centres are operating with inadequate inventory management systems of in-kind donations, which result in ineffective and inefficient resource distribution. This problem originates from using tracking and management methods that are outdated, error-prone, and unsecure such as spreadsheets, paper-based systems, and manual operation and procurement processes (Shapiro, Oystrick, Liu, & Vadlamudi, 2024). These operations often lack standardisation, have poor visibility, and lack management of unsolicited in-kind donations as current management practices aren't properly designed to govern diverse inventory, resulting in significant delays of aid that affect multiple parties including the organisation and staff, donors, and the recipients of aid (Islam, 2013).

As a result of reliance on such inadequate methods of inventory management, there are notable operational issues that hinder the organisation's ability to operate efficiently and distribute donations effectively. Delays are caused because of the dependence on traditional procurement processes of goods which affects the recipients from receiving the aid they need because of the delay of providing the critical and essential items they require. Furthermore, organisations tend to

lack up-to-date information about where and how the donations are used resulting in poor visibility and affecting their reputation, and their relationship with donors: making donors unwilling from the lack of transparency and exclusion from the donation process (Islam, 2013). And, this lack of visibility creates a significant reputational risk because of the ineffective operations that waste resources and hinders the organisation's accountability to their donors. Finally, the volunteers and staff are affected by this problem because the lack of a central digitalised inventory management system burdens them with manual, time-consuming tasks of manual data entry and organisation, categorisation, and record management that take time away from them to do more meaningful work.

Ultimately, this problem prevents donation centres from achieving their objectives, such as maximising their resources and ensuring their operations are transparent, but most importantly meeting the unmet needs of the vulnerable population. The most alarming consequence of this problem is not meeting the needs of the vulnerable, logistical chaos and poor inventory management result in shortages of critical goods and hinder the ability to quickly respond to urgent needs, compromising any donation centre's fundamental and humanitarian objective. Furthermore, this problem weakens the trust between donors and the organisation, jeopardising the centre's sustainability and expansion. Finally, this problem makes organisations unable to collect, evaluate, and analyse their inventory data, which leads to wastage of resources and leaves them unable to refine their operations so that they can maximise resources to their fullest potential.

1.3 Aim/General Objective

To create a web-based application to manage inventory for local community donation centres that struggle with inadequate inventory management systems for in-kind donations, to enable optimal resource distribution, maximise aid, and establish full transparency for stakeholders.

1.4 Specific Objectives

i. To investigate obstacles encountered by local community donation centres in managing non-monetary donations.
ii. To review the challenges of current donation inventory management solutions.
iii. To design a web-based donation inventory management system.
iv. To develop a web-based donation inventory management system.
v. To test the developed system.

1.5 Justification

This project will develop an extensive solution to solve the critical inventory management problems affecting local community donation centres by offering an optimised and effective digital solution to handle inventory.

Creating a system that provides clear and accurate records increases the trust and engagement of donors (Islam, 2013). Furthermore, it greatly reduces the time-consuming administrative work of managing inventory for the staff and volunteers, minimising their burden and reducing human error. This allows them to focus on more important and productive tasks (Ndei & Mutuku, 2021), which pushes the centre towards its goals and targets set. More importantly, this solution will greatly benefit the vulnerable population as it provides a faster and more reliable response process to their essential needs, ensuring to avoid delays and stock shortages are which negatively affect their well-being.

This solution will allow donation centres across the globe to adopt similar systems, which improves their operations, but also alleviates the burden of public social services and will complement government efforts in social welfare. As such, a solution provides comprehensive data and verifiable statistics that can be used by governments for better public policy and strategic planning for social development (Holloway, 2025).

1.6 Scope

The proposed study primarily focuses on developing an efficient and effective web-based inventory management system tailored for local community donation centres. The aim of this solution is to improve the optimality and accountability of in-kind goods inventory by strictly focusing on the digital logging and tracking of items throughout their lifecycle in the inventory, maintaining secure records of both current and past inventory items, and generating simple reports. The project focuses on three distinct stakeholders (Admin, Volunteers/Staff, Donors) providing each with specialised functions and abilities through the implementation of modern technological stack including: HTML, CSS, Javascript to create a responsive front-end. PHP will develop the logical back-end and the API communication layer which will interface with the database hosted on XAMPP, which will serve as the database that securely stores item records, inventory history, and user authentication details. Additionally, this project integrates QR codes to provide a unique

digital fingerprint for all the items. This project operates on the assumption that the donations have been sorted beforehand, checking for and removing unsolicited or inappropriate items and handling items designated for sale separately. Moreover, this project does not include any distribution logistics and will not include advanced features such as real-time SMS notifications, payment gateway integration for monetary donations, advanced predictive analytics, integration with external inventory systems, or mobile application development.

1.7 Limitations and Delimitations

Numerous challenges must be considered and solved during this project to ensure that a suitable solution is developed in the end. A major challenge is technological inexperience, especially when integrating QR code generation, display, and scanning in the solution. To handle this, a learning and training plan needs to be implemented with a focus on self-study and peer reviews to ensure understanding and solve issues early on in development. Additionally, poor communication and coordination are another challenge because they play an important role when the code development is split: poor communication leads to issues in integration because the modules were developed in isolation. To tackle this, assignment and tracking of development tasks are necessary, as they guarantee continuous progress, alignment, and identification of integration issues early on. Finally, there is the temptation to add new features beyond the scope, which can lead to a system with incomplete functionalities that affects the project's quality. This can be solved by precisely following the scope, with any alteration requests reviewed and rejected if they fall under the current exclusions in the scope.

Chapter 2: Literature Review

2.1 Introduction

This literature review demonstrates the core problem of inefficient and insufficient management of in-kind donations by local community centres. The reliance on fragmented, manual, and outdated operations leads to waste of resources and degrades the transparency of the organisation. Many existing systems, including: Humanitarian Enterprise Logistics Manager (HELM), PlanStreet software, and WASP Inventory Cloud software, try to address these issues using digitalisation and automation to centralise inventory, speed the distribution of items, and improve the quality of data. However, these systems lack important functionalities needed by centres managing in-kind donations; moreover, they impose financial and customisation constraints on organisations that implement their solutions. As a result, this study highlights the need for a specialised, cost-effective digital solution that can manage inventory effectively and connect all the parties involved in the process.

2.2 Obstacles Encountered by Local Community Donation Centres in Managing Non-Monetary Donations

Humanitarian aid operations face difficult logistical and supply chain management because of the operational challenges in managing in-kind donations. This includes the following processes: donation intake, dispatch and distribution, and reporting and accountability. The core issue this study focuses on is the management of in-kind donation inventory, encompassing the complete lifecycle of donated goods, from their arrival to distribution.

2.2.1 Reliance on Inadequate, Manual Systems

The current processes are inadequate due to their reliance on Microsoft Excel files, paper-based methods, and other manual management options. This technological gap exists because humanitarian organisations rarely integrate information technology that can manage operations in complex environments which face resource insecurity and unpredictable demand. With only a small percentage of organisations having access to enterprise resource planning software, and a lack of such software will promote manual data entry, which greatly hinders the data quality (Sigala, Kettinger, & Wakolbinger, 2020).

2.2.2 Strategic Requirements and Organisational Deficiency

Humanitarian organisations must strive to be agile, rapidly responding to short-term changes, such as recipients' needs. They must be adaptable, adjusting to different humanitarian environments, and they must have alignment by integrating and coordinating with all stakeholders involved; these dimensions ensure efficient and effective operations in the organisation (Sigala, Kettinger, & Wakolbinger, 2020). Moreover, donation centres need to strategically operate by incorporating environmental, social, and economic factors into their supply chain operations to ensure that the management of in-kind donations are flexible and sustainable (Holloway, 2025). Meeting both operational and strategic requirements is crucial for donation centres because it determines its ability to meet its targets, as well as its overall success and longevity.

2.2.3 Consequences of Insufficient Systems

The reliance on insufficient, manual systems hinders an organisation's resource distribution and transparency, because they promote poor data quality that causes errors, which shift the organisation's attention towards continuous inventory inaccuracy management rather than the strategic management using the supply chain (Sigala, Kettinger, & Wakolbinger, 2020). The inability to attain clean and accurate inventory compromises the project's goal because it prevents accountability and enables resource wastage. This highlights the lack of system integration that forces organisations to trade off between the speed of distribution and accurate inventory management (Sigala, Kettinger, & Wakolbinger, 2020). Moreover, there are non-technical barriers, such as: organisational opposition, financial burdens, and complex legalities that restrict an organisation's strategic investment of a sufficient, integrated system (Holloway, 2025), making it hard to accomplish all the necessary supply chain dimensions (agility, adaptability, alignment) that are vital for an organisation's humanitarian response (Sigala, Kettinger, & Wakolbinger, 2020). The ideal system should capture these dimensions so as to conform to regulations and cut costs (Holloway, 2025); however, this area lacks the technical ability to meet these qualities.

2.3 Challenges of Current Donation Inventory Management Solutions

2.3.1 Reliance on Manual, Error-Prone Processes

Many donation centres rely on paper-based systems, spreadsheets, and outdated software that cannot provide real-time data capture or system integration (Azman, 2024). These processes operate on manual entry of data and physical counting, using a large amount of staff time that

could be well spent on more important activities such as donor engagement, or sorting and distributing items. Additionally, these systems are subject to human errors that result in inaccurate records and delays, increasing the inefficiency in inventory management.

2.3.2 Risk of Perishable Item Wastage

The lack of real-time tracking leaves perishable items like food and medicine susceptible to spoilage or expiry, leading to a waste of resources and less items reaching the one who need them (Akkerman, Buisman, Cruijssen, Leeuw, & Haijema, 2023).

2.3.3 Lack of Integration Between Systems

A critical issue in these donation systems is the lack of integration between their systems as they are separated with various functions such as donation recording, inventory reporting or tracking. The absence of this intergeneration causes delays in decision making and resource allocation further resulting in inaccurate reporting and analysis as well as inefficiency. Ultimately, this impacts donor relationships as inconsistent data can lead to poor service and communication thus reducing transparency with donors resulting in a potential decrease in future donations (Jayadi, 2025).

However, as technology has advanced, so have inventory management systems with many now moving to become more digitalised and automated. Many digital companies have built software's to help inventory management like WMS which has helped in better tracking of inventory, reduced errors and help avoid wastage (Irere, 2023). Innovative solutions such as e procurement have been introduced in inventory management for non-profit organisations that involves digital documentation and categorising of items and tracking the items lifecycle thus improving the performance and efficiency of the inventory system as well as minimising errors (Ndei & Mutuku, 2021).

Nonetheless, gaps persist, with many solutions disengaged with donors, which is a missed opportunity to build stronger relationships and transparency with donors (Lin & Hsiung, 2024). Additionally, complex user interfaces require in-depth training, which incurs costs and reduces adoption speed, leading to inaccurate recordings and reports. Data decentralisation is another issue that prevents the creation of consolidated reports, which are necessary to increase transparency and optimal operations. These shortcomings need to be addressed to further advance the effectiveness and efficiency of donation inventory management.

2.4 Review of Existing Systems

2.4.1 Humanitarian Enterprise Logistics Manager (HELM)

The Humanitarian Enterprise Logistics Manager (HELM) is a specialised, modular enterprise system built to tackle obstacles of logistics and accountability faced by non-profit organisations, particularly those involved in disaster assistance (HELM - Humanitarian Software, 2025). It tries to solve the inherent inefficiency, lack of specific handling, and large accountability gaps that are prominent when managing in-kind donations, using an integrated solution that combines supply chain management and warehouse management systems with enterprise resource planning features, so that it can meet the complex needs of in-kind donation logistics (HELM - Humanitarian Software, 2025). HELM is a highly relevant solution reviewed in this project because it is designed precisely to cater for the non-standard processes of receiving, valuing, and tracking donation inventory, which most commercial procurement systems fail to handle.

HELM is a Software as a Service (SaaS) solution that was developed using technologies including PHP, jQuery, HTML, and CSS, as well as supporting database back-end tools like Oracle and Microsoft SQL to create a multi-layered, web-based system. Its system adopts a separated architecture with the HELM RESTful web service API used to seamlessly integrate and connect the different systems together. Moreover, it uses the Portessa Portal Builder to create secure client portals, as well as to develop a dedicated mobile app that incorporates the same properties and functions from the web-application using the Portessa web service API, ensuring accessibility to all external users like donors and staff (HELM - Humanitarian Software, 2025).

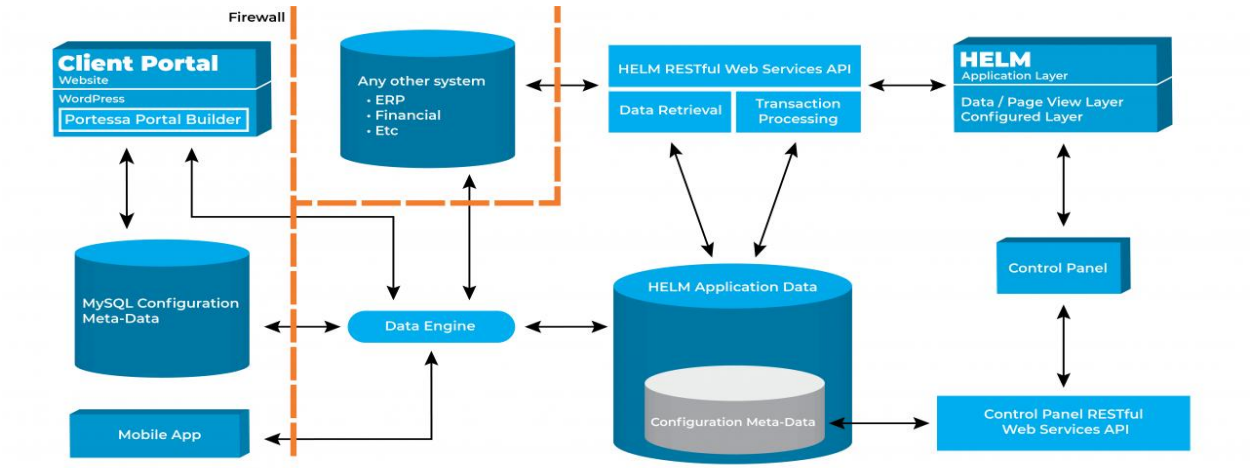


Figure 2.1 Application Architecture of the Humanitarian Enterprise Logistics Manager (HELM), (HELM - Humanitarian Software, 2025)

HELM's supply chain management system integrates numerous modules, such as the Inventory Control module, which is the central hub because of its robust features that allow access to over 100 inventory parameters (HELM - Humanitarian Software, 2025). This is crucial for items like pharmaceuticals that need tracking of lot numbers, restriction codes, and expiration dates. It includes major non-profit specific functionalities for processing in-kind donations, recognising the difference between this process and traditional purchase and receive cycles, as in-kind donations require distinct processes that include compliance requirements and can manage donor-imposed restrictions (HELM - Humanitarian Software, 2025).

The warehouse management system provides vast configurable, adaptive tools specifically for disaster assistance logistics (HELM - Humanitarian Software, 2025). A core feature is the use of techniques like radio frequency processing through a barcoded warehouse module, which enables instant data validation and real-time database updates for item movements, such a feature provides the speed and accuracy necessary for disaster responses, with timely identification and distribution of specific items saving beneficiaries lives. Furthermore, this system provides other features such as the Potential Item Master Catalog, which allows the system to automatically set up and recognise new donated items which increases the efficiency and accuracy when receiving donated goods (HELM - Humanitarian Software, 2025).

The enterprise resource planning functionality is achieved through a partnership with Unit4 Business Software and it provides a solution to financial logistics (HELM - Humanitarian

Software, 2025). It allows for the integration of logistical data and financial data (using financial and administrative tools from Unit4): The logistical data is the physical inventory data, more specifically it determines what the item is and where it is stored in the warehouse, as well as when it was distributed. The financial data provides information on the value of items, tracks donor-imposed restrictions on items, and monitors the use of the items against specified budgets and funds (HELM - Humanitarian Software, 2025). This is essential to ensure accountability and transparency, as the integrated systems can present all intakes and spending related to a project to report back efficiently and to make sure resources are used optimally.

HELM aligns with growing standards in the humanitarian sectors because its features are clearly designed to meet the sector's logistical and operational needs, its design is progressive and validated by academic and industry researchers. HELM's modular and browser-based structure adapts to different crisis situations, and it incorporates features that align with best practices for humanitarian logistics that can differentiate between commercial and aid transactions when handling donations (Saïah, Vega, & Kovács, 2023). The system is able to address sustainability challenges faced in in-kind donation management because of its real-time data collection, such fast and accurate data response is an essential foundation in advanced technologies. Pushing the industry to progress to advanced data analysis, with HELM supporting this growth by ensuring its data can enable artificial intelligence and predictive analytics to predict unpredictable in-kind donation volumes and optimise stock rotation based on expiration dates, so as to drastically reduce waste and ensure effective utilisation of donated goods (HELM - Humanitarian Software, 2025).

HELM's technological sophistication is a double-edged sword: even though its core warehouse management system is highly effective, its reliance on its barcoded warehouse module and real-time radio frequency processing creates a drastic barrier to entry and use for smaller, local community donation centres. As the dependence on specialised scanning equipment, constant power, and trained personnel becomes highly impractical or infeasible in a low resource or disaster affected environment, where manual or low technical solutions are often utilised. The system's sophistication level is advantageous to larger organisation, but is too complex and technological in smaller scale operations. Moreover, HELM focuses on providing analytical financial reports to donors rather than offering interactive or dedicated portals to build donor relationships. HELM

provides the perfect solution for larger scale operations, where the technological resources are readily available, however it is not ideal for dynamic, resource-constrained local operations.

2.4.2 PlanStreet

PlanStreet's Inventory Management Software aims to solve the problem of inefficient inventory management of donated goods by replacing the fragmented and error-prone manual records, spreadsheets, and isolated systems with a centralised, digitalised, and automated solution (Inventory Management Software, 2025). PlanStreet's software targets core problems such as inventory inaccuracies, complex reporting, and delays in distributing critical items.

The software organises and tracks the entire inventory lifecycle using multiple features and techniques. It creates QR codes for all goods in stock, allowing staff to scan them and assign donations to recipients, providing real-time transaction processing by enabling quick and error-free scanning during distribution to link donations to specific client case files. Staff can use modifiable forms and data validation for donation intakes, capturing important details like category, description, quantity, and expiration date, and it provides up-to-date visibility of stock levels using a cloud-based system. Additionally, it eliminates manual, paper-based burdens by automatically updating stock and creating records of distributed items, increasing the accuracy and efficiency in operations (Inventory Management Software, 2025).

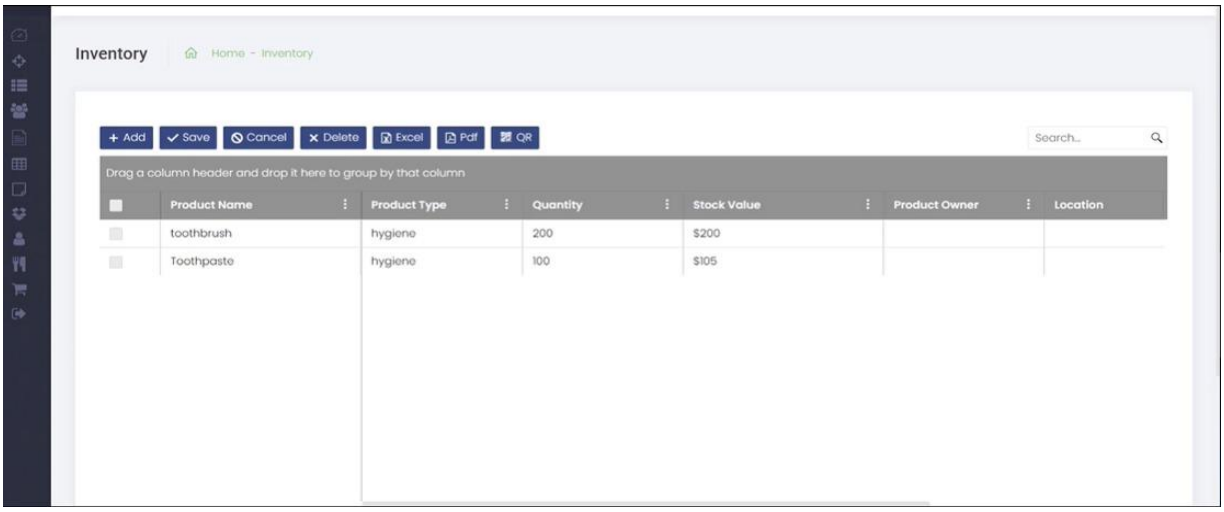


Figure 2.2 PlanStreet's Inventory Management System, (Inventory Management Software, 2025)

PlanStreet offers enterprise-level features using modular, cloud-based software that can handle the precision and specificity of in-kind donation inventory. It enables transparency by generating reports that showcase the use of the donations, building the donors' trust. Moreover, it uses Microsoft Power Business Intelligence to convert inventory data into useful information that allows organisations to apply more analytical trends that enhance their operations. This application directly tackles issues of limited reporting and allows the organisation to present the actual impact of the donations (Inventory Management Software, 2025).

Despite its strengths, PlanStreet's use of highly complex emerging technology (Business Intelligence Analytics) poses a challenge for local centres that heavily rely on volunteers, who require simple and inherent interfaces to minimise errors and training expenditures. Furthermore, the implementation costs and the customisation requirements are other challenges because of the financial constraints of small donation centres. The case studies from PlanStreet indicate that successful adoption and implementation of their solutions was not immediate as there was the need of project management including data relocation and extensive training sessions for users across multiple organisations. For instance, the need to tailor the system to align with WHEDco's unique program structures and reporting needs, indicates that the solution is not a simple and ready-to-use solution (Inventory Management Software, 2025). The need for customisation and training is another financial and time burden, which essentially excludes smaller, more resource-constrained community centres from utilising the platform's benefits for meeting the operational needs.

2.4.3 WASP Inventory Cloud

The WASP Inventory Cloud solution is designed to solve the problem of inefficient inventory management of in-kind donations within non-profit organisations that operate with time and budget constraints, which leads to wastage and ineffective operations. The solution provides a centralised platform for monitoring and managing inventory, guaranteeing accuracy in all inventory records and minimising the time spent on administrative tasks (Non Profit Inventory Management Made Simple, 2025).

The system is centered around barcode technology and mobile access which enables proper sorting, processing, and restocking, as well as speeds up inventory checks and reduces human errors in the system. The system offers both cloud-based or on premise inventory management,

and provides all the necessary hardware (scanner and printers) and software to run the entire system (Non Profit Inventory Management Made Simple, 2025). This allows non-profit organisations to efficiently track donations, simplify and speed up operations, as well as effectively meet their goals.

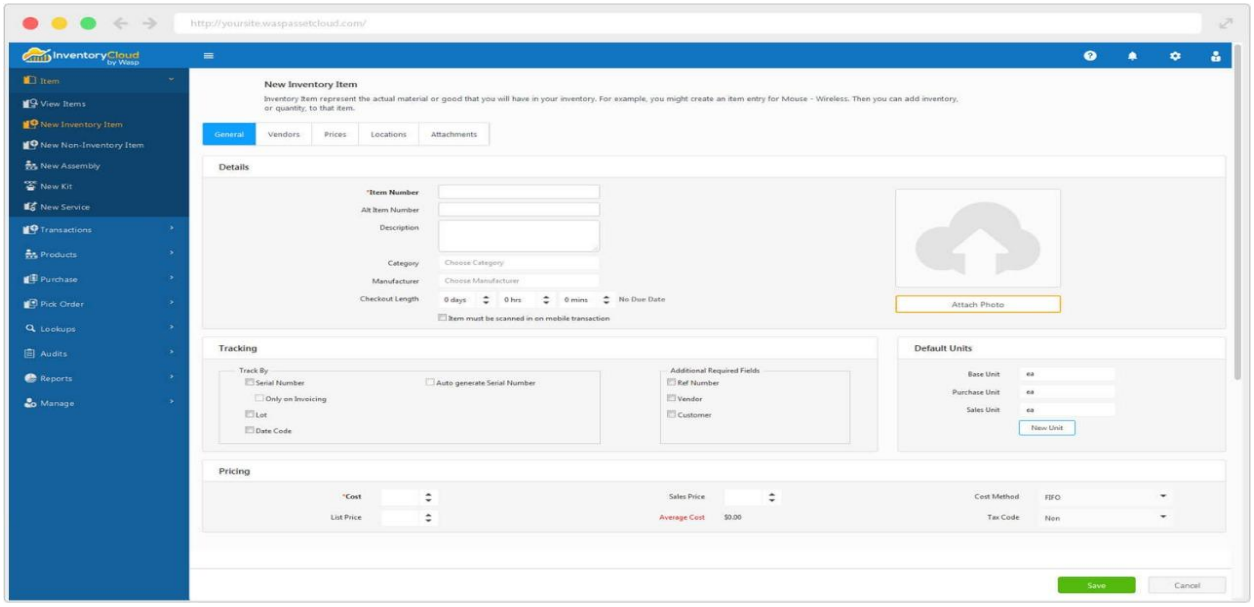


Figure 2.3 New Inventory Item on Inventory Cloud by WASP, (Non Profit Inventory Management Made Simple, 2025)

WASP Inventory Cloud solution addresses the issue by providing tools that enhance the operational efficiency and resource maximisation of an organisation, which is necessary in humanitarian logistics. It incorporates advanced features that push the industry forward, including the use of demand forecasting and automated reordering to ensure outstanding maintenance of stock levels, greatly reducing overstocking or understocking in the operations. Moreover, the system integrates a reporting hub that presents critical information like donated item categories, and donation frequency; which allows non-profits to boost their effectiveness by making data-based decisions, and increases their transparency with their donors. Furthermore, it uses real-time cloud-based tracking, which clearly benefited the Maniilaq Association by saving over 390 hours a year that were previously spent on manual data transfer: this highlights the platform's ability to save time and maximise resources, even in remote and resource-constrained environments. Finally, the system is designed to be user-friendly and simple to use, which reduces the time and financial burden of having to train staff to use the system (Non Profit Inventory Management Made Simple, 2025).

Despite its strengths, WASP's provision of the entire system (software, hardware, and consumables) instructs organisations to purchase proprietary hardware directly from WASP, which is particularly difficult for budget-constrained organisations and limits their ability to utilise existing hardware or seek cheaper alternatives. The dependence on a single vendor also affects long-term lifespan of the system, as organisations are entirely reliant on WASP for equipment updates and repairs. Over time, this dependency can result in significantly higher operational costs as organisations are unable to benefit from competitive pricing of generic hardware: such a limitation blocks the financial and technological freedom of the organisation during the lifespan of the system.

2.5 Conceptual Framework

The conceptual framework outlines the workflow of the inventory management system of in-kind donations for local community donation centres. It defines clear roles and responsibilities for all the stakeholders. The process begins with a universal Login page that authenticates all three primary actors (Donors, Staff and Admin) and grants them access to their specific modules. The donor module provides donors with information on critical items needed by the centres. It has a donation form that donors can complete with information about the donations they will make. It allows them to track their donations and the stage at which they are at. It provides them with simple, system-generated reports and lets them submit feedback on the operations. The staff module provides real-time stock visibility, allowing staff to upload information on critical understocked items that are needed. They can update the donation status of donors with the stage at which their donations are at. Additionally, it enables them to manage inventory using generation of QR codes during intake and scanning QR codes when distributing. Lastly, it allows them to view and respond to donor feedback. Finally, the admin module allows administrators to manage and oversee the entire system's functionality, operations, performance, and its users. They can generate reports on the inventory or of users in the system. All these modules are connected and performed using a standardized API layer which securely stores and retrieves data from the database, such as user profiles, inventory items, and reports. The API connects the front and back end, and it ensures reliable access to the information when it is requested by authorized users, thus establishing seamless and secure communication.

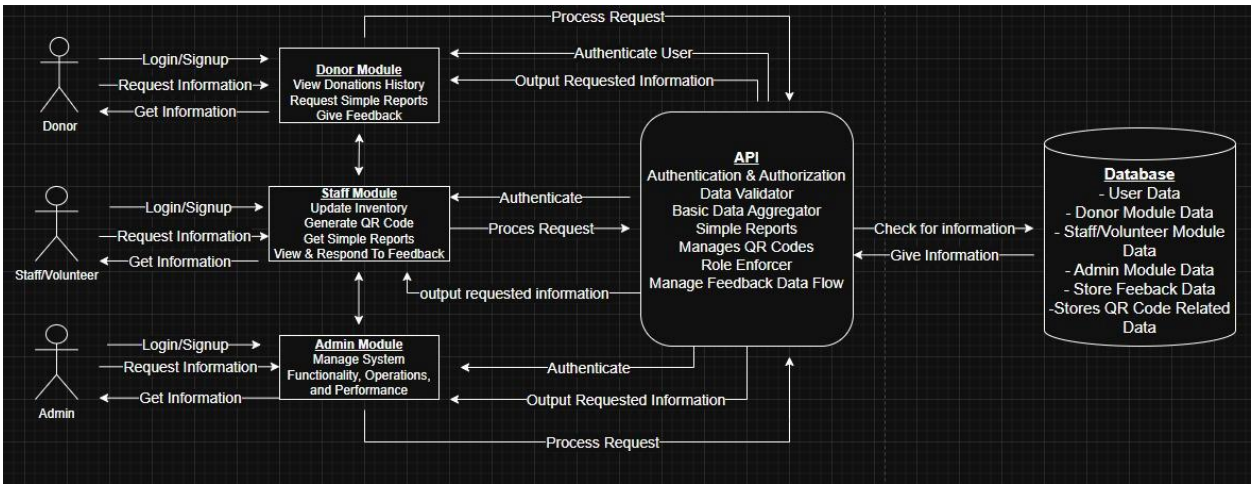


Figure 2.4 Conceptual Diagram

Chapter 3: Methodology

3.1 Introduction

This chapter entails project development of a web-based in-kind donation inventory management system using an Object-Oriented Analysis and Design (OOAD) approach and the Scrum methodology, under the Agile (Adaptive) methodology. This process involves four phases, including: Requirements Gathering to create a specified product backlog that defines the functional and non-functional requirements based on research. Planning and Design phase used OOAD diagrams like ERD's and wireframes to demonstrate interconnections between the different entities and technological infrastructures. Implementation, using Sprints to develop the solution using HTML, CSS, JavaScript, PHP, and XAMPP. Finally, testing, using Jest to automate both unit and integration testing, and manually testing whether the product will meet the specified requirements and solve the problem at hand.

3.2 Research Approach

This project undertook an Object-Oriented Analysis and Design (OOAD) approach because it offers a natural way to model the real-world entities and relationships which are inherent in inventory management system for in-kind donations. OOAD has proven to be effective for improving inventory management systems based on its different angle on problem solving (Sanmocte & Costales, 2025). Similarly, it effectively manages complexity by organising the system into interacting objects that represent donors, donations, inventory items thus making it easier to track and update inventory dynamically (M.R., et al., 2013). Moreover, inventory systems need to evolve over time and the OOAD approach allows continuous refinement and adaptation of the evolving requirements thus beneficial for this project.

3.3 Methodology

The methodology selected to develop a specialised in-kind inventory donation management system is Scrum, which is a progressive and repetitive agile software framework used to manage the development of projects (Sachdeva, 2016). This choice was made because of the unique number of constraints in the problem: while the system must handle complex inventory tracking and reporting, it must also be user-friendly, and simple enough for a short-staffed or a volunteer-run community donation centre to adopt. Scrum operates through short, focused work intervals called sprint which promotes continuous adjustment rather than rigid adherence to upfront planning

(Sachdeva, 2016). This framework is managed by key roles, such as the product owner who maintains the prioritised list of features in the product backlog, and the development team who develop the solution. To confirm progress, sprint reviews and the process itself is improved during the sprint retrospective to ensure that the solution is constantly aligned with the main goal. The overall software development process for this project will understand the solution in a structured manner because of the mapping of the adaptive activities onto the traditional lifecycle phases below.

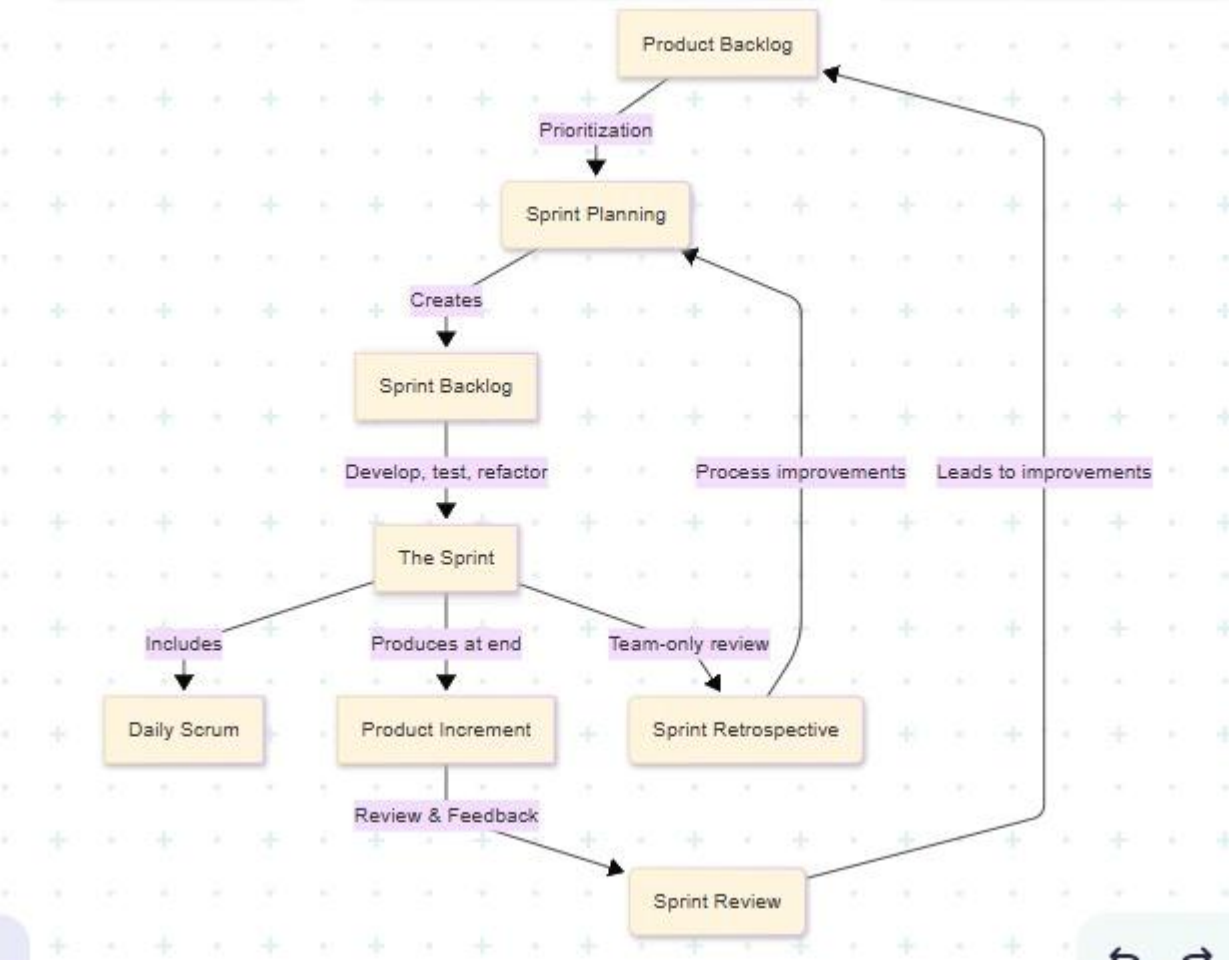


Figure 3.1 Scrum Process Mapping for Project Development

3.3.1 Foundational Phase: Requirements Gathering and Product Backlog Creation

This initial phase focused on deeply understanding the users needs and transforming the conceptual framework into specified, prioritised development tasks. To shift from the current manual, fragmented inventory systems, a specific solution that captures the workflows and roles of each

and every stakeholder is needed. Therefore, the primary activity was collection of qualitative and quantitative data through targeted secondary research, including systematic analysis of current organisations' operations to determine data and workflow requirements, as well as evaluation of existing solutions to assist in the swift determination of the functional and non-functional gaps that exist in current solutions, as well as to identify what is coveted by the target users. This information will be the basis of the product backlog: The product backlog serves as the single, prioritised source for all requirements, ensuring that features are ordered according to their functional value and immediate user need. The approach focuses on delivering essential, impactful features first, which is crucial for quick and quality development and meeting all the requirements.

3.3.2 Planning and Design Phase: Object-Oriented Design and Architectural Planning

This phase ensured that the system supports modular, and role-focused architecture whilst promoting maintainability and efficiency. The Object-Oriented Analysis and Design (OOAD) was the perfect choice because the system naturally revolves around numerous interacting objects such as inventory items, differing users, and donation records. OOAD can effectively manage the complexity of in-kind donation data because it allows the design to integrate, connect, and create relationships between these entities, providing uniformity when managing the inventory. Architectural planning was demonstrated using a set of OOAD diagrams including use case diagrams that clearly defined the system's boundaries and showcased high level functionalities; class diagrams that showcased the structure and relationships between core objects; sequence diagrams that visualised the dynamic flow of critical features and actions; and wireframes that were developed to ensure a simple and user-friendly interface is created from the beginning. The diagrams in this phase were created using draw.io, Lucidchart and many more tools because they developed visuals that align with both the solution envisioned and the agile principles defined.

3.3.3 Implementation Phase: Development Environment and Tools Utilised

The implementation phase will be executed through a sequence of sprints, where the design requirements are converted into functional software increments. The system is a web-based application because of its universal accessibility on any device (mobile phones, laptops, tablets) in order to support donation centres by cutting down the cost of procuring specialised or proprietary hardware. The system will use the following technological stack to develop a secure, three-tier architecture that separates the external level, conceptual level, and internal level: The external level

(user interfaces) will be built using HTML, CSS, and JavaScript because they enable rapid, modular development that is necessary for the sprints. The conceptual level will be handled by PHP as it unifies with other technologies using JavaScript across the front-end and back-end, eliminating complexity and pushing forward the development (Kesavan, Gay, Thevessen, Shah, & Mohan, 2023). The internal (database) layer will be managed using XAMPP because it can easily handle the changing and unpredictable nature of in-kind donation data, and its real-time synchronisation (Yadav, Dhakad, ., Panchal, & Kaur, 2024), which is crucial for staff and volunteers that need quick inventory visibility. The full development will take place within Visual Studio, an integrated development environment because it can support all the technological aspects and infrastructure to develop the solution.

3.3.4 Testing Phase: Testing the System's Quality and Validation

In the scrum methodology, testing is integrated throughout the development process, with each sprint defined as "done" once each feature has been fully tested. The solution will be constantly verified and validated against the standards and specifications set in the product backlog. The types of testing to be carried out include: unit testing, integration testing, and system testing, which: checks if the individual components (units) work correctly on their own, ensures seamless data flow between the front-end and the backend, and evaluates whether the complete software meets the specific requirements, respectively. Such immediate feedback is necessary to verify and validate the functional and non-functional requirements of the system, and to carry out the testing Jest can be used as the standard testing framework that can be used to automate both unit and integration tests (Penta, 2004). This immediate feedback sequence is needed to fulfil the scrum principle of constant inspection and improvement, allowing the developers to confirm the system's quality and quickly resolve underlying issues before moving on to the next sprint.

3.4 Deliverables

3.4.1 Documentation

The documentation deliverables for this project includes all the essential documents that detail the development stages and finalisation of the in-kind donation inventory management system for local community centres. This encompassed the comprehensive proposal that outlines the problem statement, objectives, scope, methodology, literature review, and limitations and delimitations.

3.4.2 System Design Artifacts

This included the development of visual design diagrams such as use cases, class diagrams, sequence diagrams, and entity-relationship diagrams (ERD). Additionally, wireframes of the user interface have been created, each illustrating what the users will see and interact with in the system.

3.4.3 Core Application Modules

The core deliverable is a working web-based application with the following key modules: An authentication module for user login or registration and role-based access. A donor module to submit, track, and generate simple reports on donations. A staff module to process donations and generate QR codes. And, an admin module to manage the users and the system. Additionally, robust security measure will be implemented, including password hashing and input validation. These measures ensure the integrity, safety, and protection of all data in the system and will deliver a complete and demonstrable solution to the problem of managing in-kind donations.

Chapter 4: System Analysis and Design

4.1 Introduction

This chapter entails a breakdown of the system's analysis and design, including a comprehensive understanding of the problem, the stakeholders and their obstacles, and what can be done to satisfy them. It dissects all the system's functional requirements for all users, from donors to administrators, as well as defines the non-functional requirements that will enhance the system's performance. Finally, it includes diagrams that depict the system's design, providing a visual to understand the system's layout, structure, users, and operations.

4.2 Requirement Analysis

Stakeholders are currently operating with manual, error-prone, paper-based or spreadsheet systems that result in operational failures, including resource wastage, lack of donor trust, and staff frustration. They are seeing a rise in digital inventory management solutions that offer comprehensive tools and technologies that can tackle in-kind inventory management challenges, which can solve the challenges they currently face: Donors are uninformed about their donations on aspects like what it was used for, whether it was used, and whether it was what was critically needed. Staff or Volunteers waste time and resources when managing the inventory with the current processing, which hinders them from distributing the goods to those in need. Administrators are unable to get accurate, up-to-date reports that can assist in resource allocation decisions in the centre. These current issues signify the need to shift to a proactive, real-time solution that can provide the tools to manage in-kind donations effectively and efficiently and to measure the success of our system: more specifically, core tasks and functionalities; metrics and targets are set in stone to ensure high performance and to solve the current problem.

4.2.1 Functional Requirements

4.2.1.1 Donor Functional Requirements

The donor should be allowed to input credentials for login/registration. They should be able to view and edit their user profile, i.e. allow the donor to change their username and password. The system should allow the donor to generate reports based on their data. They should be able to view the critical donation items needed by the centre. The system should allow donors to view previous donations made to the centre. The donors should be able to write feedback to the staff. Lastly, the

system should allow donors to submit information on donations made to be able to view the status of their donations (To be received', 'Processing', 'Rejected', 'In Inventory', or 'Distributed').

4.2.1.2 Staff Functional Requirements

The staff should be allowed to input credentials for login/registration. They should be able to view and edit their user profile, i.e. allow staff to change username and password. The system should allow the staff to generate reports based on their data. The staff should be able to view, add or remove donation items from the inventory in real-time. The staff should be able to categorise the donated items and add details about them such as expiry date. The system should allow staff to view and respond to donor feedback. The system should allow staff to write and post the critical in-kind donations needed for the donors to view. The staff should be able to edit the donation status of a donor's donation information with 'To be received', 'Processing', 'Rejected', 'In Inventory', or 'Distributed'. The system should allow staff to generate a QR code for a new donation item to be added to the inventory. The system should allow staff to scan a QR code to distribute the item and make its status as distributed.

4.2.1.3 Admin Functional Requirements

The admin should be allowed to input credentials for login/registration. They should be able to view and edit their user profile, i.e. allow the admin to change their username and password. The system should allow the admin to generate reports based on their data. Admin should be able to view all users and validate new staff users or new admin users. The system should also allow the admin to oversee the inventory. The admin should be able to manage users, removing users who are no longer a part of the system, as well as modify their roles, such as making an existing staff user an admin user. The admin should be able to view donor feedback and staff responses, as well as respond to feedback made by the donors.

4.2.2 Non-Functional Requirements

4.2.2.1 Security

The system must prevent unauthorised users from accessing the modules and authenticate users to their assigned roles, ensuring they can only access their designated endpoints. Manual testing will check if this functionality is upheld. Moreover, all sensitive information (including passwords and identifiable data) should be safely and securely stored, using encryption for all data transfer, and penetration testing will determine the effectiveness of the system's security. Finally, the system

must validate all user input, checking and structuring the data from the front-end before processing and saving it in the back-end to avoid malicious input and improper input that leaves the system vulnerable to damage, unauthorised access, and attacks. Security testing will be done to verify the user input before acceptance.

4.2.2.2 Capacity

The system must be able to store at least 1000 user profiles and up to 10,000 inventory records, which is possible through MySQL's high-volume storage and retrieval capabilities. Simulation testing, by importing fake inventory items into the database, will be done to determine if the system can meet these targets. Furthermore, the system should support at least 20 users simultaneously with no deterioration in its performance. Simulation testing, using automated testing tools, will be used to imitate large multiple-user sessions to verify the system's ability to perform with many users.

4.2.2.3 Portability

The system's user interface layout must adapt to numerous user devices, such as mobile phones, tablets, and desktops, to allow staff to manage inventory using their devices and not have to purchase specialised hardware for the centre. A responsive design framework will be developed, and manual testing will confirm the functionality of the user interface across different devices of varying screen sizes.

4.2.2.4 Reliability

Backups are essential for restoration in the event of errors and malfunctions, to protect and safely store the system's data. Simulations of data loss will test the system's ability to restore data from the latest export and verify its consistency and integrity. Additionally, data consistency must be upheld throughout the system's front and back end, with updates or changes made in a module reflected across other modules and the database within 5 seconds. Integration testing will determine whether the system's units can properly interact and update data on time.

4.2.2.5 Maintainability

A modular approach and the enforcement of separate code development will ensure that each module is independent and able to meet its requirements with no hindrance to the entire system. Unit testing will check that each module is fully operational alone before integration.

4.2.2.6 Scalability

The system must handle a sudden increase in users interacting with the system, managing resources based on the number of users, as well as guaranteeing availability and preventing system crashes. Simulation of user increase can be done to test the system's ability to handle growing use.

4.2.2.7 Usability

The system's user interfaces should allow users to complete core tasks with no training; by creating a minimalistic user interface that focuses on simple, understandable, and efficient operations. Unit testing will verify that each module meets the defined requirements and checks whether the completion rate of core tasks in each module is 100%. In addition, this system should handle errors by providing clear and comprehensible error messages that guide users through failed procedures, letting them fix the issue with ease. For instance, inputting a negative quantity in the inventory is incorrect, and a message should pop up to let the staff user know that "the quantity field cannot be negative". Manual testing will be done to check if there are adequate error messages in place in the system.

4.2.2.8 Performance

The system's transaction speed is as follows: QR code generation and display should take 10 seconds. Updates between the front-end user interfaces and the back-end should take 5 seconds. Simple report generation should take less than 5 seconds. The time taken to perform these critical tasks is pertinent to boosting the speed of operations and performance testing, using developer tools, will confirm the time taken to perform these tasks against their targets.

4.3 Diagrams

4.3.1 Use Case Diagram



Figure 4.1 Use Case Diagram

4.3.2 Class Diagram

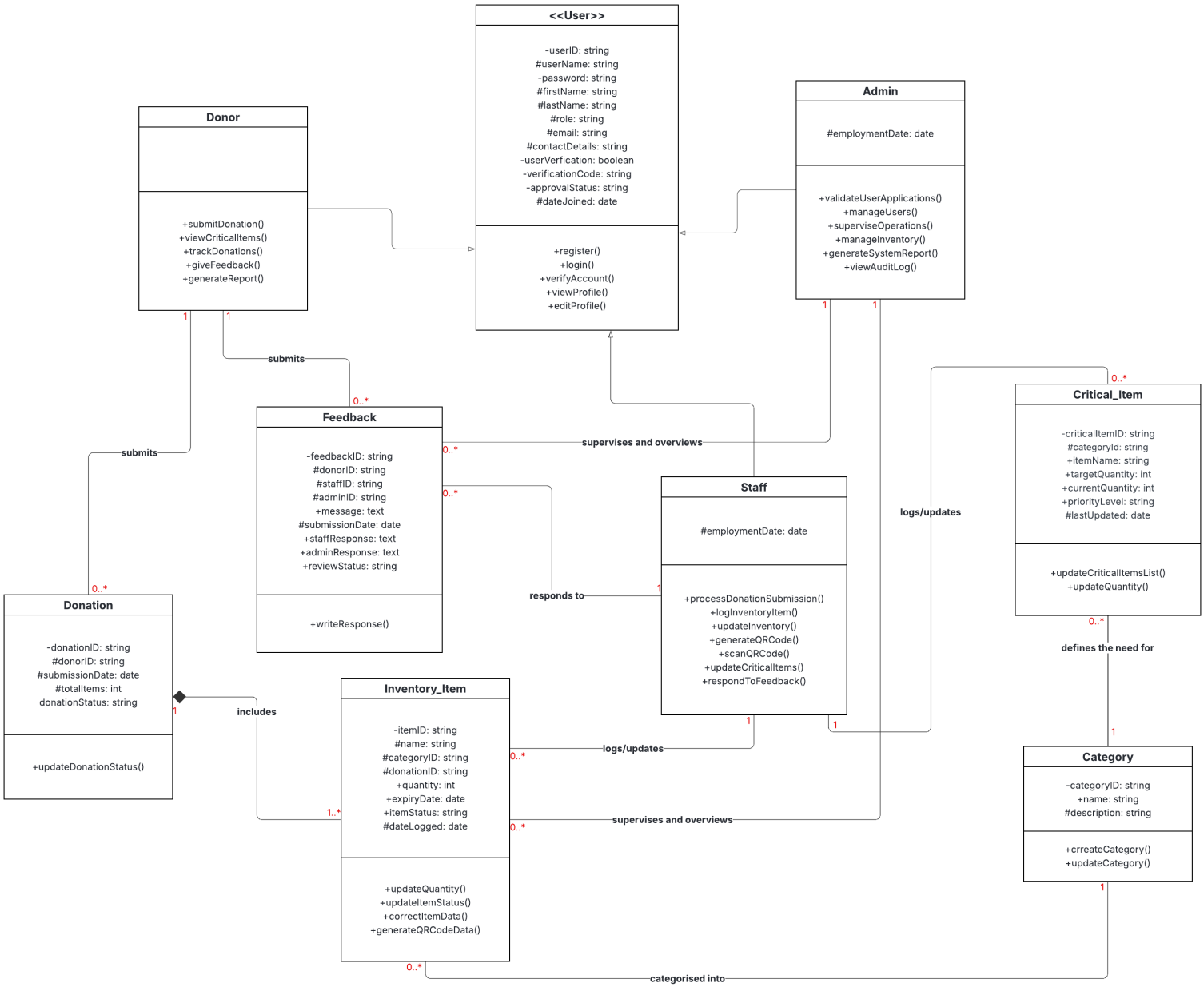


Figure 4.2 Class Diagram

4.3.3 Sequence Diagrams

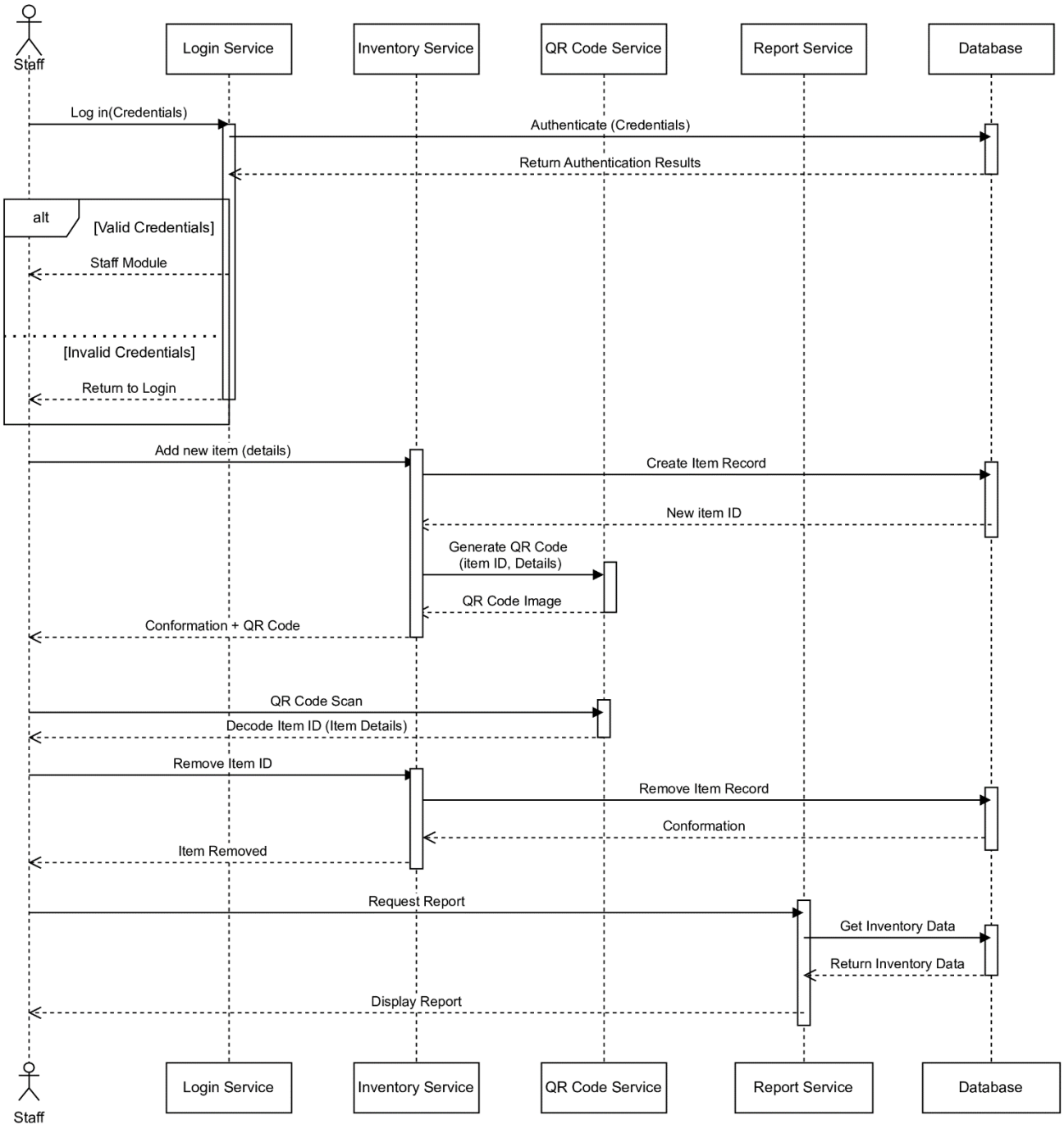


Figure 4.3 Staff Sequence Diagram

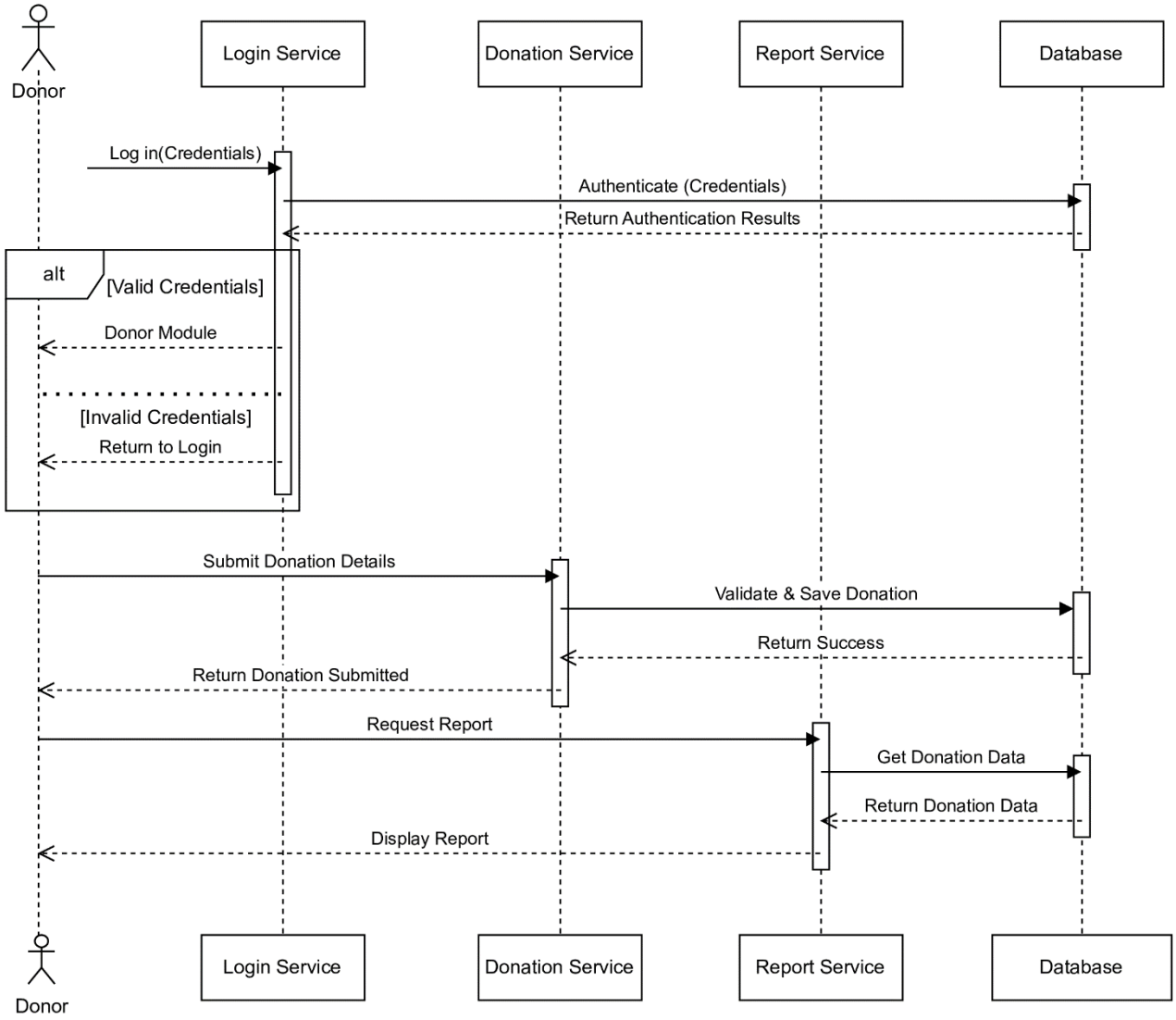


Figure 4.4 Donor Sequence Diagram

4.3.4 Entity Relationship Diagram

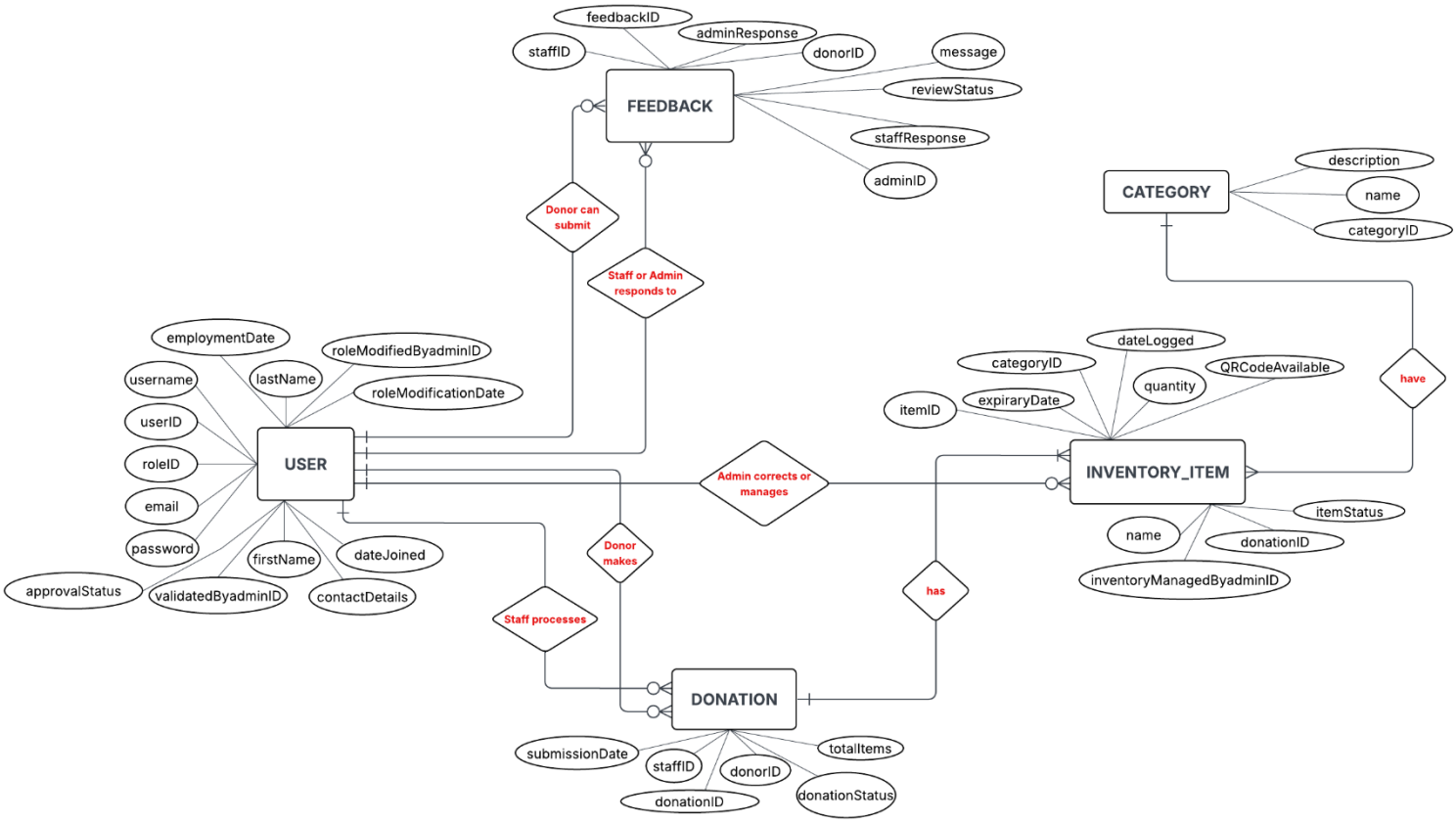


Figure 4.5 Entity Relationship Diagram

4.3.5 Database Schema

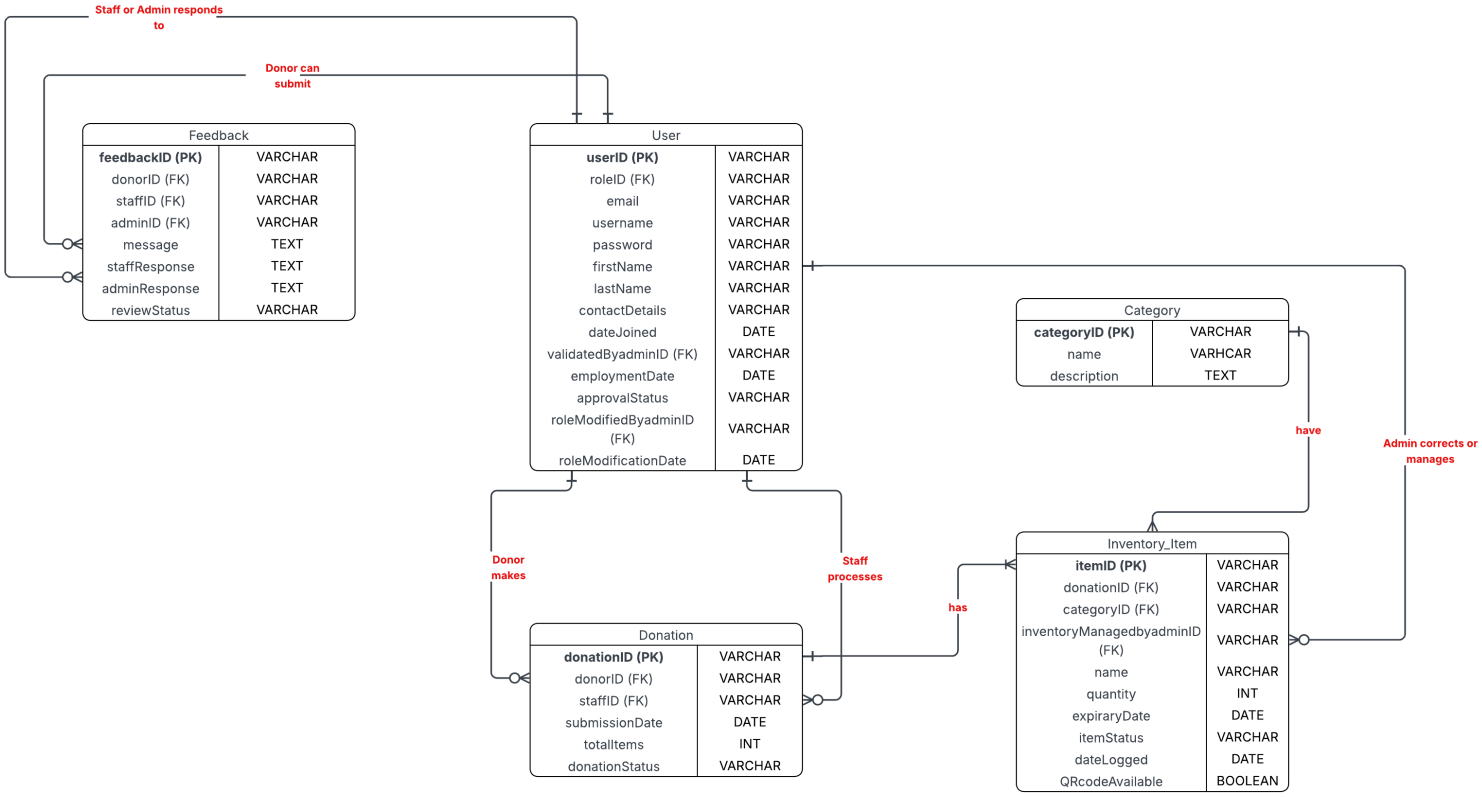


Figure 4.6 Database Schema

4.3.6 Wireframes

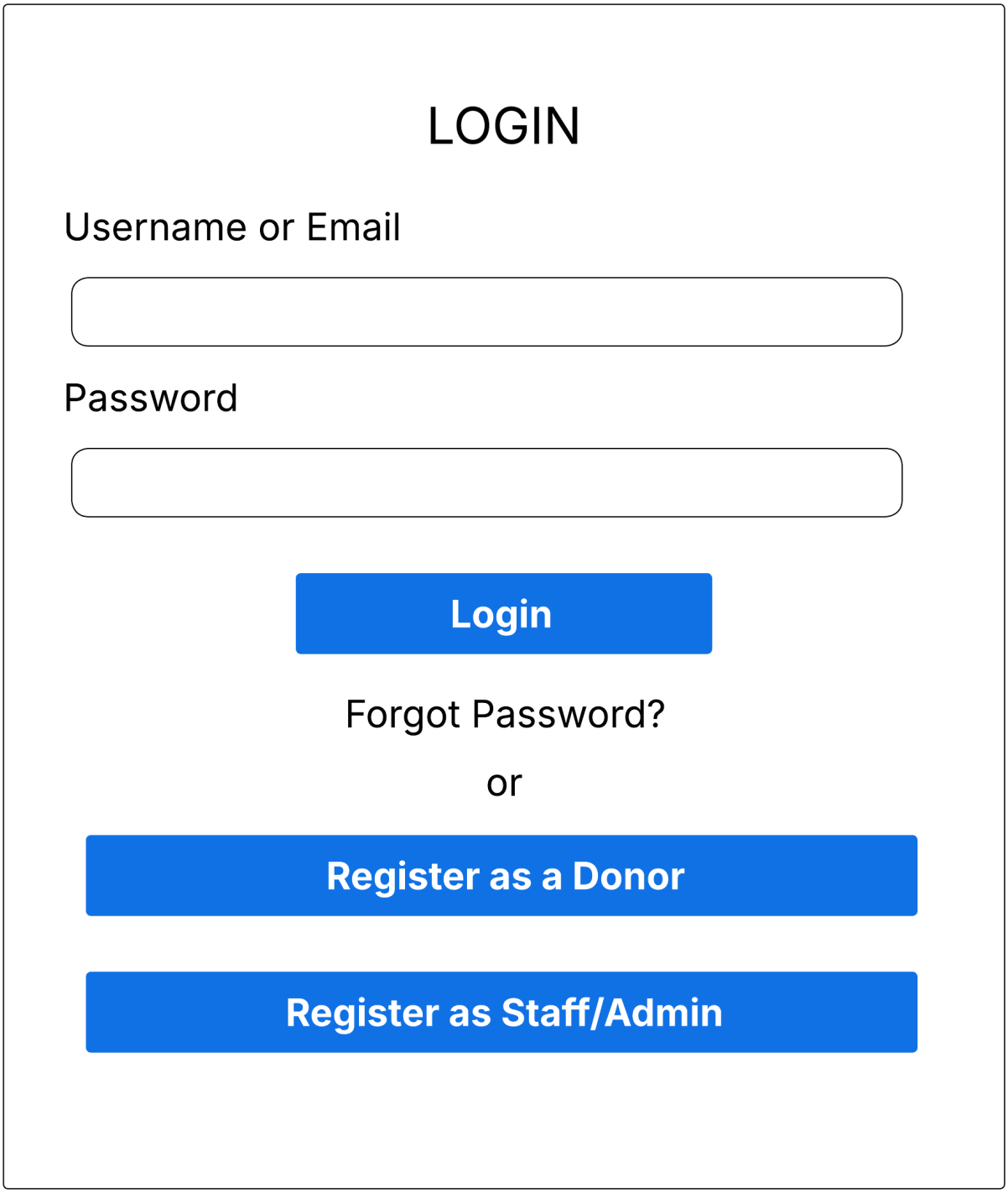


Figure 4.7 Login Page Wireframe

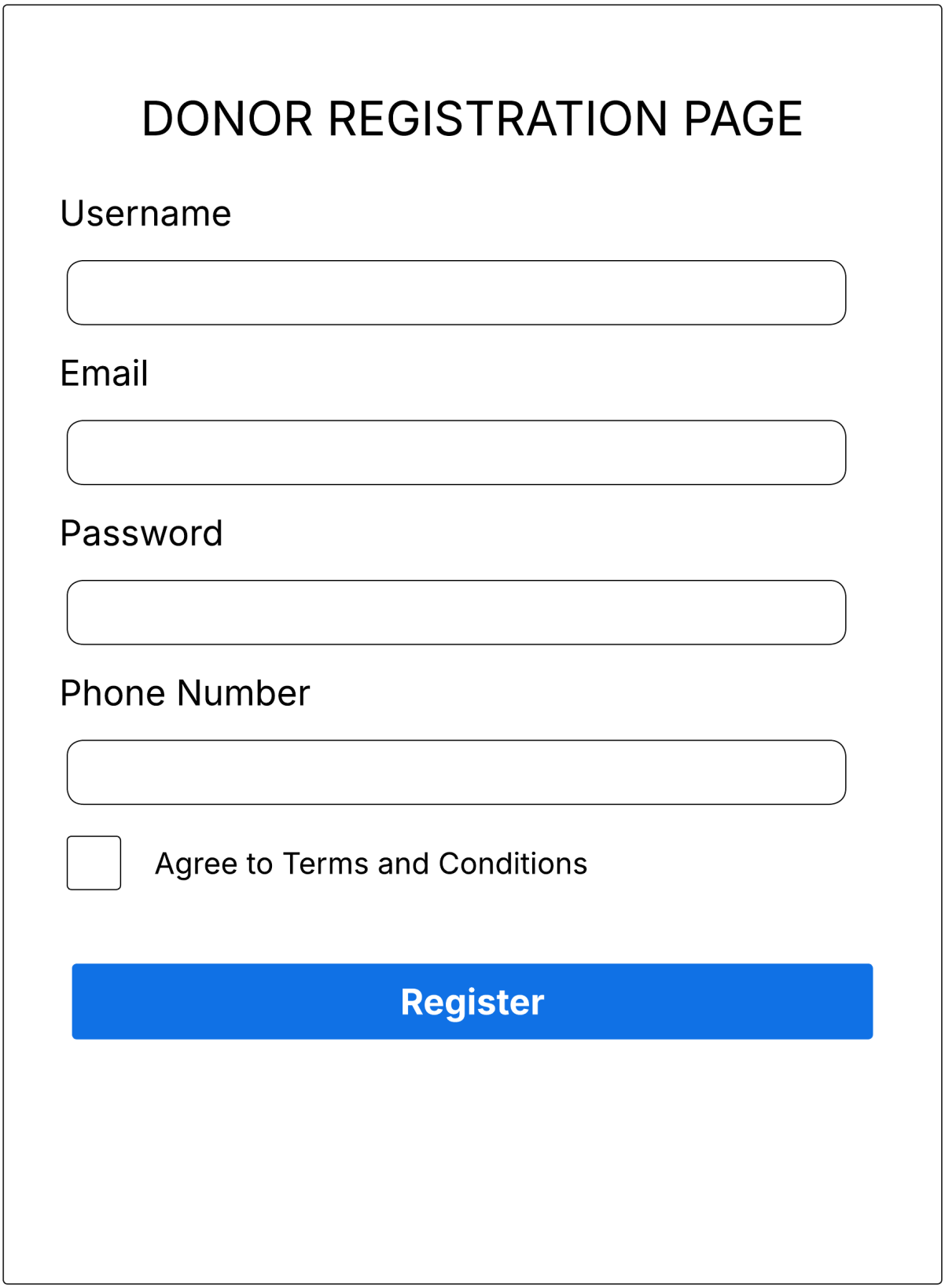


Figure 4.8 Donor Registration Page Wireframe

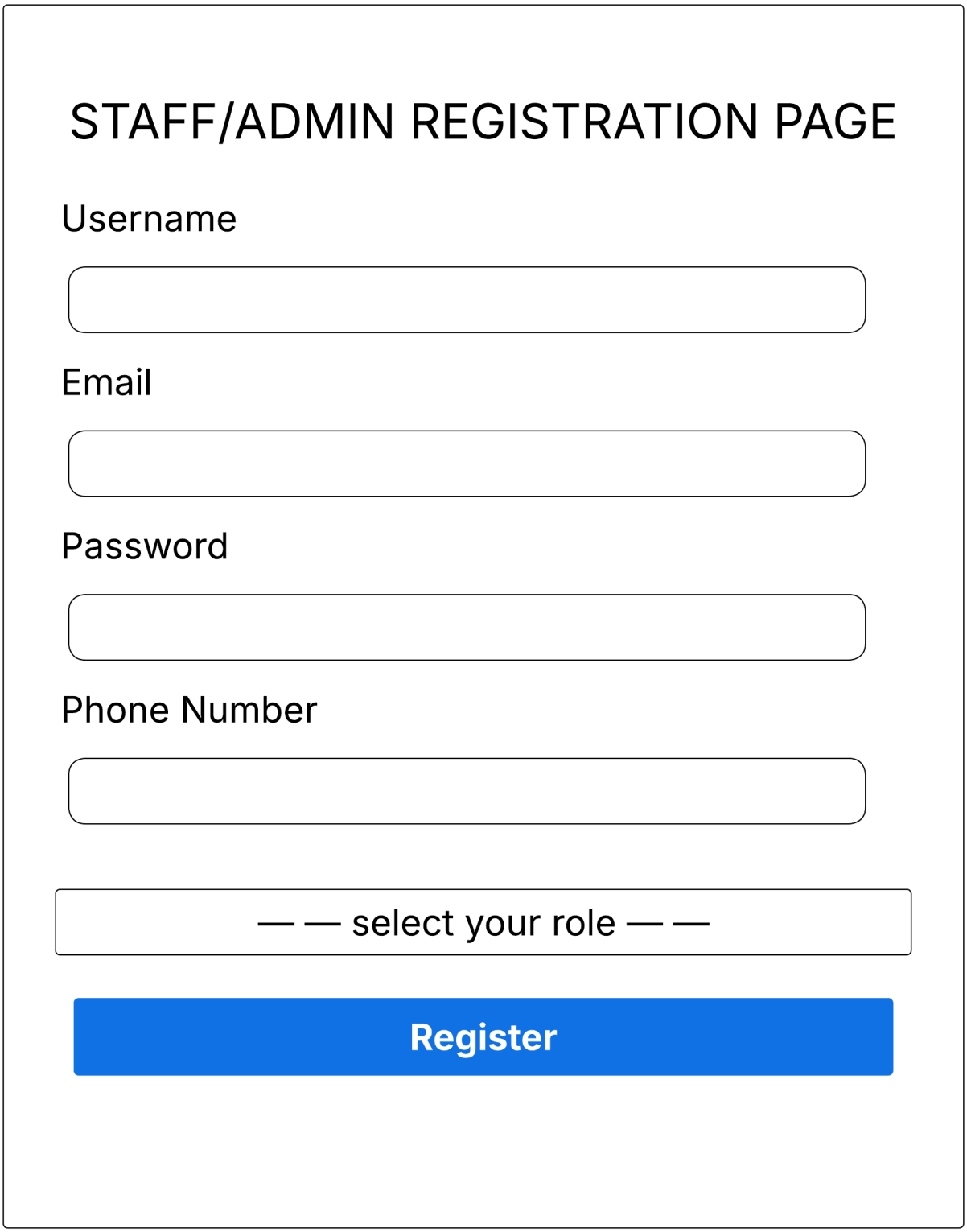


Figure 4.9 Staff/Admin Registration Page Wireframe

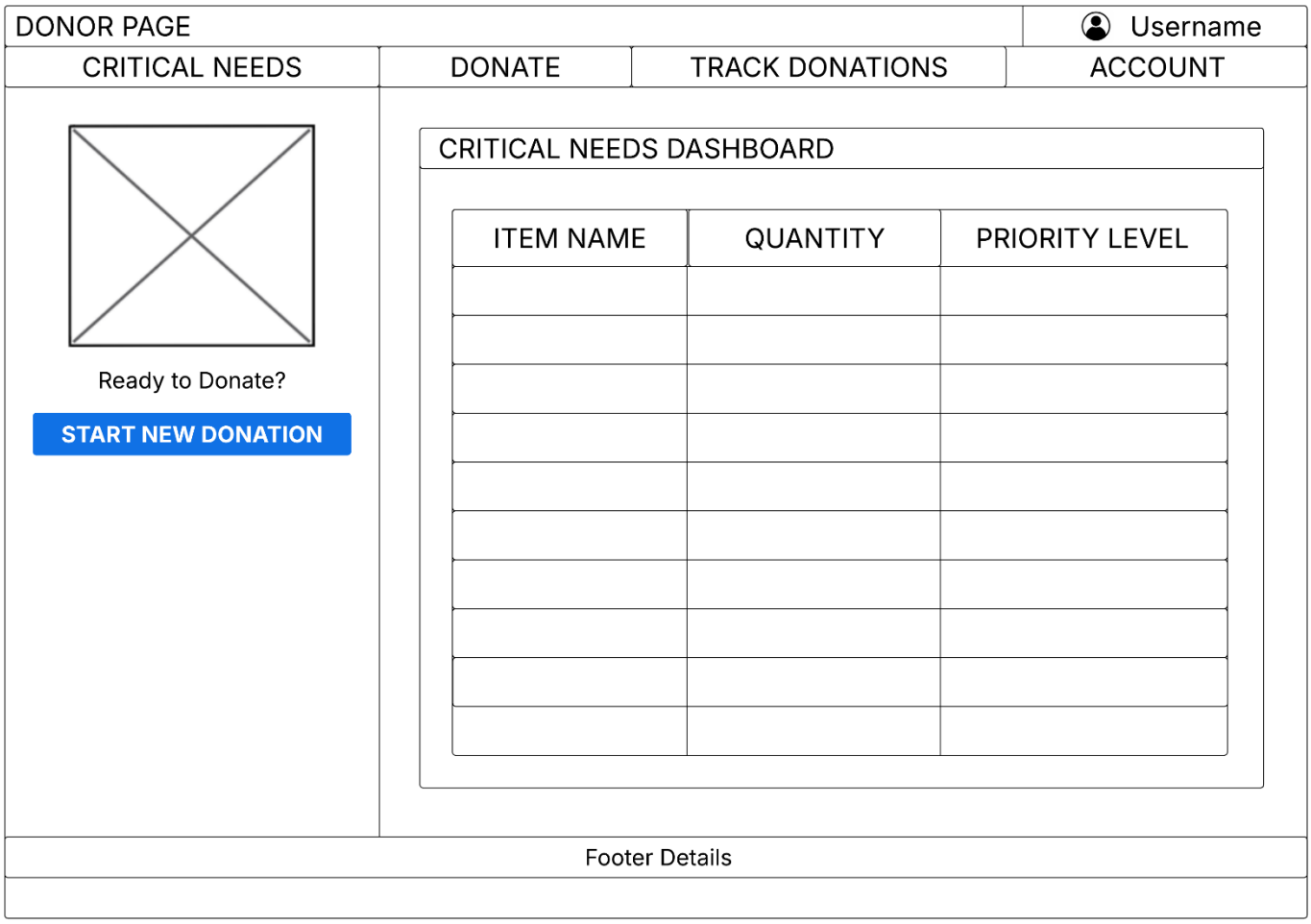


Figure 4.10 Donor Page Wireframe

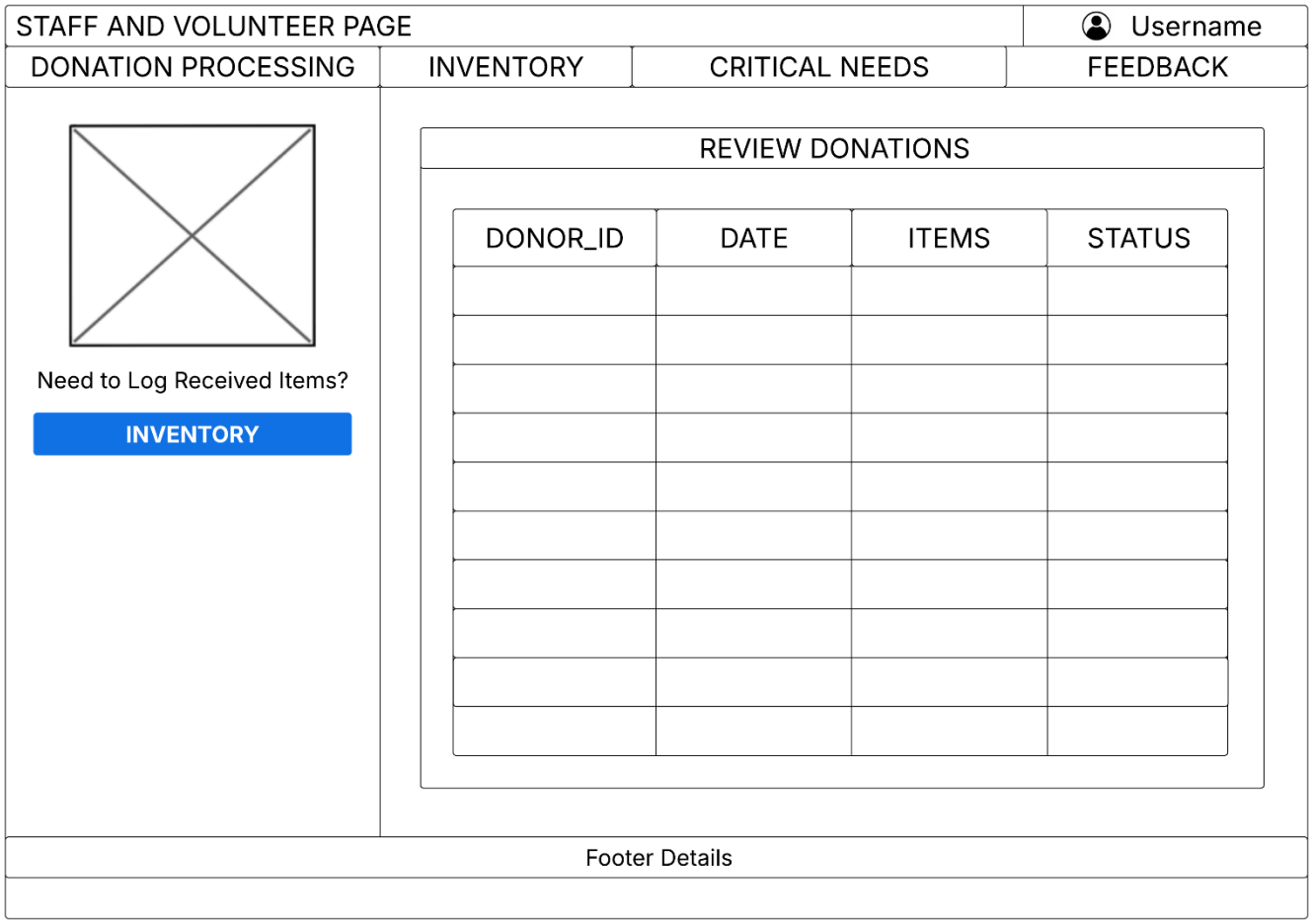


Figure 4.11 Staff and Volunteer Page Wireframe

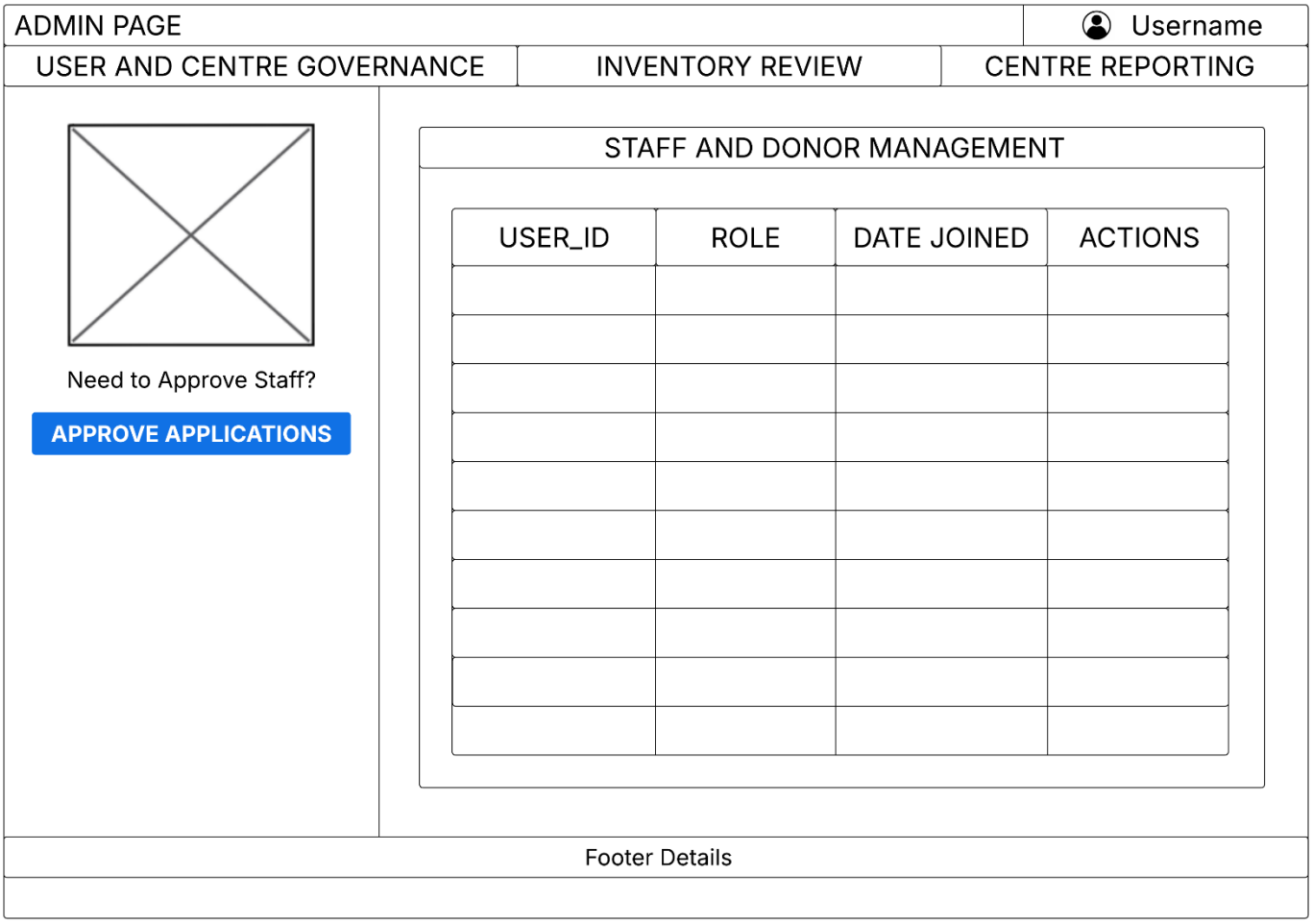


Figure 4.12 Admin Page Wireframe

Chapter 5: Development and Testing

5.1 Introduction

This chapter provides a comprehensive breakdown of the system's development and testing phases. It details the frontend and backend construction, including the tools used. Additionally, it outlines role-based workflows for donors, staff and admin. It discusses stakeholder interactions by outlining error messages users may encounter and how they interact with the requirements. Furthermore, it acknowledges deviations from the proposed implementation and discusses the challenges faced during the development such as database query errors, session management issues, and QR code integration. Finally, it includes detailed testing for each module which features specific test cases and their results supported by sample appendices.

5.2 Setup Description

5.2.1 Solution Architecture

The front-end of the solution has been built using HTML, CSS, and JavaScript. It was built using Visual Studio which is an integrated development environment as it supports all the aspects and infrastructure needed to develop the frontend. Similarly, backend was primarily developed using PHP. The application runs in a XAMPP environment that also provides an Apache server to host and execute the PHP scripts. Additionally, the database is also hosted in MYSQL of the XAMPP environment. XAMPP was used as it combines PHP, Apache and MySQL which made it ideal for development and testing.

The email verification utilised EmailJS which uses three unique identifiers including the public key, and service and template IDs. EmailJS works in JavaScript and uses the public key to initialise and determine the user the email is being sent from. And, once a user submits their registration form, the data is sent to the php file server, and it validates the account and provides it a unique verification token and then the server returns a JSON (JavaScript Object Notation) that contains the user's email, username, and verification token which is then processed by the client-side and if the registration is successful, the construction of the verification URL is done including the email and token with "encodeURIComponent". The email is then sent using the service and template IDs with objects mapping the respective placeholder for the user's information including their username, email, and verification link and then the email is sent if the process is successful, if not it catches the error informing the user that their verification email wasn't sent.

The QR code in the Staff Inventory uses php and the phpqrcode library to link the inventory items to the QR code. When the staff member clicks on "Generate QR" button, the JavaScript capture the click and retrieves the inventory_id of the item and sends it over to the php file handling code generations. This file then constructs a URL using the device's IP as a server IP, and it includes the path to the next php file which is the "QRView.php", for instance: http://{serverIP}/DIMS_FINAL/System/QRView.php?id={inventory_id}. It uses the library files and the QRcode::png() function to convert the URL to the QR code, saving them on the dedicated folder and then the Javascript code updates the database, saving the image path into the QR code path column of the specific item and it is displayed next to the inventory item and can be downloaded [See Appendix i].

The design of the solution is simple for easier navigation and less expenses to be spent on training how to use the application.

5.2.2 Test Environment

The environment used to test the system is as follows:

Table 5.1 Test Environment Table

Operating System	Windows 11
Processor	11th Gen Intel(R) Core (TM) i7-1185G7 @ 3.00GHz (3.00 GHz)
RAM	16.0 GB (15.7 GB usable)
IDE	Visual Studio Code [Version 1.106.3]
Hosted On	XAMPP [Version 8.2.12] • Apache Server [Apache/2.4.58 (Win64) OpenSSL/3.1.3 PHP/8.2.12] • Database client version: libmysql - mysqlnd 8.2.12
Web Browser	Microsoft Edge and Chrome

5.2.3 Operational Workflow

5.2.3.1 Donor

A donor begins with registering an account. Upon registering, they will receive a verification email to confirm their identity so that they may login. Once verified, they can now login and be redirected to the dashboard of the donor module. This will provide an overview of key module features and allow donors to submit and view responses to their feedback. Furthermore, they can access a sidebar navigation menu for direct access to other pages of the module. Donors can visit the critical needs page that highlights the items needed the most by the centres to make it easier for donors to decide on what to donate. To donate the donors will navigate to the "Donate" page to complete a detailed donation form. This form will allow them to list up to 10 items with a maximum quantity of 1000. Donors can also track their donations status from the "Track Donations" page. This allows them to see what stage their donations are at, such as "To be Received", "Processing", "Rejected", "In Inventory" or "Distributed". Donors can generate reports for individual donations or a comprehensive report for all their donations. Lastly, the settings page allows donors to update their username, phone number, or password.

5.2.3.2 Staff

Staff can register and once approved by an admin they can log in and will be redirected to the dashboard of the staff module. Furthermore, they have a side bar navigation with links to different pages. They have the critical needs page where they can update what kind of in-kind donations the centre requires or what is low in stock for donors to see. A donations page to see the donations made by donors and update the status of the donors' donations. They can view what is in the inventory which have been organized categorically. Staff are also able to add items to the inventory easily and generate QR codes for category item. This reduces the manual administrative burden on the staff saving time and making the process of inventory management more efficient. Additionally, unlike spreadsheets being open to everyone this system has a login page thus only authorized users have access to the inventory. Staff are also able to reply to feedback gotten from donors and can give feedback to admin. The staff are also able to simple generate reports which lists items in inventory or items that have been distributed. Lastly, settings page provides staff the ability to change their username, phone number, or password.

5.2.3.3 Administrator/Admin

A new admin must register and then be approved by an existing admin before the account becomes active. Once approved, the admin can login and be redirected to the dashboard that provides a real-time overview of platform activity, displays key metrics, recent donations and new user activity. Other module functions are accessed through the sidebar navigation menu such as "Manage Users". This is where admins can approve, reject or remove users from the system. Admins can generate reports on staff/admin users, donor users, or a comprehensive report of all users. They can oversee the inventory to verify proper functionality and data accuracy. Furthermore, admins can view and respond to feedback submitted by donors and see the replies given by staff. Lastly, the settings page provides admin the ability to update their own username, phone number, or password.

5.2.4 Stakeholder Interaction

5.2.4.1 Errors

When an unauthorized staff/admin tries to login they will be get an error message [See Appendix 1]. If a donor tries to exceed over 1000 quantities, they will get an error message [See Appendix 2].

5.2.4.2 Functional Requirements

All users can login and be redirected to their respective modules via the login page. All users can generate reports based on their modules.

Once donor's login, they can view critical donation items needed by the centre from the critical needs page. They can also submit donations via the donation page. Additionally, donors can view and track donations made on the track donations page. Donors can give feedback to staff/admin. [See Donor Test Case Results].

Once staff login, they can view, edit or delete critical donation items on the critical needs page. They can also view donations made by donors and update the donations' status. The staff can view, add or remove items from the inventory from the inventory page. Staff can reply to feedback given by donors. [See Staff Test Case Results].

Once admin login, they can manage users on the page called 'Manage Users'. They can approve or reject users. Admin can also change the role of users from staff to admin or admin to staff.

Additionally, they can view all the donors in the system and remove any user from the system. Admin can overview inventory. Admin can view, respond or delete any feedback. [See Admin Test Case Results].

5.2.4.3 Non-Functional Requirements

Email Verification after registration which is a security measure used to help prevent false registrations, secure personal data and prevent fraudulent behaviours thus keeping the system secure. Furthermore, manual testing was done to make sure the system prevents unauthorised access to the modules which can be seen in the 5.3. [See Appendix C]

Each unit was tested to confirm smooth operations and to check if all requirements were met. Moreover, manual testing was done to check if error messages were working properly i.e. maximum quantity, unauthorised login errors, etc. [See Error Message Appendices].

Manual Testing was done to test the user interface on different devices such as a phone, laptop and tablet. [See Appendix 3].

5.2.4.4 Report Generations

Donors can generate reports listing the items they have donated along with their donation status (To be Received, Processing, etc) [See Appendix EB]. This allows donors to monitor their contributions on their module without having to constantly ask updates from the centres and provides immediate proof of their donations being received and logged by the centre. Thus, fostering transparency and building trust between the donors and centres.

Staff can generate reports listing items in inventory or items that have been distributed. This enables staff to make informed decisions about resource allocation or identify urgent needs. [See Appendix L]

Admins can generate individual (admin/staff or donor) user reports or a comprehensive report on all users in the system. This serves as a critical tool for oversight and decision making and aid in user management [See Appendix P].

5.2.5 Deviations from Proposed Implementations

While the core functionality of the system was achieved, several features were simplified or deferred during the development and implementation. First, simple reports were meant to be

generated based on user data; however, tabular printouts were implemented instead using FPDF. These documents aren't considered analytic reports as they don't manipulate the data to capture and present insights to the users, including showing trends of the most items that fall under a category, or showing charts and other pictorials that present the central tendency of the data, for instance.

Next, the system should allow the staff to scan QR codes to update existing items' status and details; however, the QR code function implemented only deals with marking an item as distributed. This limits the QR code from serving as a data management tool that maintains items' lifecycles, which significantly reduces its utility for the distribution process. However, this simplification provides staff with a gain in efficiency, as they can immediately scan and distribute items when in storage, without needing to go back and forth to a desktop or manually record the details, which will speed up the process and ensure accuracy.

Finally, full inventory management entails being able to update/edit item details at any point while in inventory. However, the inventory section solely focuses on managing items that are in inventory and are to be distributed; with all the edits and updates made on the donation management page, and once set to "in inventory", it is stored and awaits distribution, with no further editing allowed. The benefit of this approach, however, is that it creates separate duties in each of the pages: staff can fully focus on determining the quality and suitability of the item before adding it to the inventory, while the inventory section is created to perform rapid and efficient distribution.

5.2.6 Challenges

One of the biggest challenges faced was ensuring that all the SQL queries matched the database accurately. The constant errors when running the files were a result of misnaming the different fields, and because the system is heavily reliant on inserting, extracting, and displaying information from the database, it was a challenge that really hindered the development of the system.

Another challenge faced was managing the redirections and the session expiration whilst interacting with a web page. The session would expire once the user attempted to interact with an action, it would fail to properly redirect the user, leading to the session code at the top to execute and redirect the user back to the login page. Handling QR codes and storing their paths in the

database was difficult because it entailed creating multiple php and JavaScript files that had to be interconnected and ensure that the transfer of all information of the QR code was accurate. The main difficulty was in creating a path that could connect one device to another (server to client), because it required that, from the point of generation that both of the files are connected to the same network to ensure that it works.

Moreover, integrating the correct files from the php library was difficult, with any difficulty stopping the QR code generation process. The synchronisation between the Staff Donation Management and the Staff Inventory, and ensuring that once a donation item is in inventory, it can no longer be edited on the donation side, and can only be further processed in the inventory. It was difficult because most of the code worked on changing the items' status, and constant alterations on both sides became confusing and hectic to handle.

Finally, generating the PDFs using the FPDF library was difficult because it required precise positioning of all the content, with every element having to be placed using specific coordinates. This made it difficult to handle the data in the table, because it needed to be well-spaced and placed in the borders of the table columns. Additionally, since generating tables with a long list of data and from the database, there was a lot of complex code and logic that had to be applied to ensure that it would display all the information, and not overlap the content.

5.3 Testing

Testing was done manually for each module and some of their features to ensure functionality.

5.3.1 Test Cases

5.3.1.1 Donor Module Test Cases

Table 5.2 Donor Module Test Cases

Test Case ID	Test Type	Functionality Tested	Description	Test Environment	Test Tool	Test Data
1	Security	Login Form Input Validation	Verifies if correct credentials have been inputted to be redirected to donor module, else	Web Browser (Microsoft Edge)	Manual	Username: Juls Password: Password123#

			prevents authorization			
2	Maintainability & Usability	Registration	Check if user is added into database	Web Browser (Microsoft Edge) & MYSQL	Manual	Username: Juls Email: nahjulbatulk@gmail.com Password: Password123# Phone: 0706055671
3	Performance	Registration Email Verification	To see if donor users can receive the email verification to verify their account	Web Browser (Microsoft Edge) & Email	Manual + Automated Email Verification Sender	Username: Juls Email: nahjulbatulk@gmail.com Password: Password123# Phone: 0706055671
4	Usability	Donation Form submission	Checks if donors can properly submit donation form & if data is in database	Web Browser (Microsoft Edge) & MYSQL	Manual	Item Name: Blanket Quantity: 10 Description: For the winter. Item Name: Jackets Quantity: 10 Description: To keep them warm.
5	Usability	Donation Tracking	Checks if after donors submit	Web Browser	Manual	Item Name: Blanket

			donation forms if they can track it	(Microsoft Edge)		Quantity: 10 Description: For the winter. Item Name: Jackets Quantity: 10 Description: To keep them warm.
6	Usability	Feedback	Checks if donor can submit feedback and view staff/admins reply	Web Browser (Microsoft Edge)	Manual	Donor Feedback: This System is Amazing! Staff Feedback: Thank You so much. We are glad it's working for you :)

5.3.1.2 Staff Module Test Cases

Table 5.3 Staff Module Test Cases

Test Case ID	**Test Type**	**Functionality Tested**	**Description**	**Test Environment**	**Test Tool**	**Test Data**
7	Security	Login Form Input Validation	Verifies if correct credentials have	Web Browser (Microsoft Edge)	Manual	Username:StaffUser Password: Staff@Password123

			been inputted to be redirected to staff module, else would prevent authorization.			
8	Security	Authorization of Staff users to log in	Login Form only accepts user if registration request was accepted by admin. User here was rejected to see if they can still log in	Web Browser (Microsoft Edge)	Manual	Username: Julsisstaff Password: Password123#
9	Usability & Maintainability	Critical Needs	If critical items is added it should reflect in the database	Web Browser (Microsoft Edge) & XAMPP MYSQL (database_dims)	Manual	Perishable / Non-perishable: Non-perishable Category: office/stationary supplies Item name: Books Priority: Low
10	Usability & Maintainability	Critical Need deletion	Checks if a critical need once deleted is also removed from the database	Web Browser (Microsoft Edge) XAMPP MYQL (database_dims)	Manual	Item Name: Books Quantity Needed: 200 Priority: Low

11	Usability & Maintainability	Donation status update	Checks if staff can update the donors' donations status	Web Browser (Microsoft Edge)	Manual	Changing Update Item Status from 'To be received' **to** 'Processing' for Donation 9 for **Items:** Blankets Quantity:10 Description: For the winter.
12	Usability & Maintainability	When Donation Status is 'In Inventory'	When Donation Status is 'In Inventory' the items should be in Inventory.	Web Browser (Microsoft Edge)	Manual	Item Name: Jackets Quantity: 10 Category: Clothing Perishable: No Update Status: 'In Inventory'
13	Usability & Maintainability	Expiry Date Alert Box	When expiry date is a date that is passed there will be an alert box	Web Browser (Microsoft Edge)	Manual	Current Date: 3/12/2025 Past Date: 11/11/2025
14	Usability & Maintainability	Negative Quantity Error	System does not allow negative quantity items	Web Browser (Microsoft Edge)	Manual	Donation #12 Quantity: -1
15	Usability & Maintainability	Once Donation status is 'Distributed'	Once donation status is 'Distributed' it cannot be changed	Web Browser (Microsoft Edge)	Manual	Donation #10 Donor: Juls Total Items: 2 Items: Books – distributed

						Bananas - distributed
16	Performance	Report Generation	Staff can generate module reports in less than 5 seconds	Web Browser (Microsoft Edge)	Manual	Data of the inventory
17	Usability & Maintainability	Feedback Test	To check if staff can see feedback from donors & respond to them as well.	Web Browser (Microsoft Edge)	Manual	Donor Feedback: This System is Amazing! Staff Feedback: Thank You so much. We are glad it's working for you :)

5.3.1.3 Admin Module Test Cases

Table 5.4 Admin Module Test Cases

Test Case ID	**Test Type**	**Functionality Tested**	**Description**	**Test Environment**	**Test Tool**	**Test Data**
18	Security	Login Form Input Validation	Verifies if correct credentials have been inputted to be redirected to staff module, else prevents authorization	Web Browser (Microsoft Edge)	Manual	Username: AdminUser Password: Password123#

19	Security	Authorization of admin users to log in	Login Form only accepts user if registration request was accepted by admin. User here was rejected to see if they can still log in	Web Browser (Microsoft Edge)	Manual	Username: John Password: Johnisadmin123#
20	Usability/ Maintainability	Approve/Reject Users	Check if admin can approve or reject users	Web Browser (Microsoft Edge)	Manual	Users Jane (Staff) and Adam(Admin)'s account will be pending
21	Usability	Role Change	See if admin can change the role of a user from staff to admin or admin to staff	Web Browser (Microsoft Edge)	Manual	Change User Mark from Admin to Staff
22		View or remove donor users	Check if admin can remove donor users from the system	Web Browser (Microsoft Edge)	Manual	User: imanisuser will be deleted.
23	Performance	Generate Reports	Check if admin can generate reports in less than 5 seconds	Web Browser (Microsoft Edge)	Manual	Demo data of current users in the system
24	Usability	Feedback Overview	Admin should be able to View and respond to donor and view staff responses	Web Browser (Microsoft Edge)	Manual	Donor Feedback: This System is Amazing! Staff Feedback: Thank You so

						much. We are glad it's working for you :) Admin Feedback: Thank You For your Feedback!

5.3.1.4 Systems Test Cases

Table 5.5 Systems Test Cases

Test Case ID	Test Type	Functionality Tested	Description	Test Environment	Test Tool	Test Data
25	Performance & Usability	QR Code Generation	If QR codes can be generated for items to be added into the inventory	Web Browser (Chrome)	Manual	Item Name: tissue Quantity: 1
26	Performance & Usability	QR Code Scanning	If QR codes can be scanned to remove items from the inventory	Web Browser (Chrome) + Phone Camera	Manual	Item Name: tissue Quantity: 1
27	Security & Usability	Email Verification	If users can get email verification	Web Browser (Microsoft Edge)	Manual + Automated Email Verification Sender	Username: Juls Email: nahjulbatulk@gmail.com Password: Password123#

						Phone: 070605567 1

5.3.2 Test Case Results

5.3.2.1 Donor Module Test Case Results

Table 5.6 Donor Module Test Case Results

Test Case ID	Expected Results	Actual Results	Status	Appendix
1	The system should allow the authorized user to login	The system does allow the authorized donor user to login	Pass	[See Appendix A]
2	Once a donor user register it should reflect in the database	Registered donor users are shown in database	Pass	[See Appendix B]
3	Once a donor registers it should send them an email to verify their account	Registered donor can receive the email to verify their account	Pass	[See Appendix C]
4	Donation Form should submit successfully when all requirements of the for is met and should reflect in the data in the database	The donation form does submit successfully when all requirements of the for is met and does reflects in the data in the database	Pass	[See Appendix D]
5	Once a donation form is submitted, the donors can track the specific donation made	The donors can track the specific donation once the donation form is submitted	Pass	[See Appendix E]

6	Donor should be able to submit a feedback and see a staffs reply	The donor can submit a feedback and see staffs reply	Pass	[See Appendix EA]

5.3.2.2 Staff Module Test Case Results

Table 5.7 Staff Module Test Case Results

Test Case ID	Expected Results	Actual Results	Status	Appendix
7	Once an authorized staff logs in they should be redirected to the staff dashboard	The authorized staff was successfully redirected to the staff dashboard	Pass	[See Appendix F]
8	A staff that registered and was rejected should not be able to log in	The rejected staff 'Julsisstaff' could not log in	Pass	[See Appendix G]
9	When Staff adds an item to the critical items list will it reflect in the database	Item added to critical items list by staff is successfully reflected in the database	Pass	[See Appendix H]
10	Once staff deletes a critical item from the critical items list will it be removed from database as well	Once staff deleted a critical item from the critical items list it was also removed from the database	Pass	[See Appendix I]
11	Can a staff be able to edit the donation status of a donation made by donor	Staff can successfully edit the donation status of donations made by donor	Pass	[See Appendix J]
12	If a staff puts the donation status as 'In Inventory' will it reflect in the inventory as well	When a staff puts the donation status as 'In Inventory' it does reflect in the inventory as well	Pass	[See Appendix K]

13	When a staff puts the expiry date as a date that has already past it should give an alert box.	When a staff puts the expiry date as a date that has already past it does give an alert box	Pass	[See Appendix KA]
14	Staff should not be able to put a negative quantity for items	Staff cannot put a negative quantity for items	Pass	[See Appendix KB]
15	An item that is marked as distributed as its donation status cannot be changed again	Once an item is marked as distributed its status cannot be changed again	Pass	[See Appendix KC]
16	Staff should be able to generate inventory reports in less than 5 seconds	Staff can generate inventory reports in less than 5 seconds	Pass	[See Appendix L]
17	Staff should be able to view and reply to feedback given by donors	Staff can view and reply to feedback given by donors	Pass	[See Appendix M]

5.3.2.3 Admin Module Test Case Results

Table 5.8 Admin Module Test Case Results

Test Case ID	**Expected Results**	**Actual Results**	**Status**	**Appendix**
18	Once an authorized admin logs in it should redirect them to the admin module/dashboard	An authorized admin can successfully login and be redirected to admin module/dashboard	Pass	[See Appendix N]
19	The system should not allow an unauthorized rejected users to login	The unauthorized rejected user could not login	Pass	[See Appendix NA]

20	An admin should be able to see all pending user approval requests	Admin can see all pending user approval requests	Pass	[See Appendix O]
21	Admin should be able to change the role of a user	Admin can change the role of a user	Pass	[See Appendix P]
22	An admin should be able to delete a user	Admin was able to delete a user	Pass	[See Appendix Q]
23	Admin should be able to generate a report on users	Admin can generate user reports	Pass	[See Appendix R]
24	Admin should be able to view and reply to feedback	Admin can reply and view feedback	Pass	[See Appendix S]

5.3.2.4 System Test Case Results

Table 5.9 System Test Case Results

Test Case ID	**Expected Results**	**Actual Results**	**Status**	**Appendix**
25	A QR Code for an item should take less than 5 seconds to generate and should be able to generate	QR Code for the item tissue was able to generate and took less than 5 seconds	Pass	[Appendix T]
26	Once a code is scanned it should pop up the item for that QR Code	Once item tissues QR code was scanned the pop came up	Pass	[See Appendix U]
27	The system should send an email verification once a donor registers.	They system successfully sends an email verification once a donor registers.	Pass	[See Appendix C]

Chapter Six: Conclusions and Recommendations

6.1 Conclusions

This project seeks to solve the logistical and accountability issues present in the manual, paper-based tracking and management of in-kind donations inventory. The main aim was to create a centralised web-based application that was able to provide standard donation intake, provide real-time visibility of critical needs, feedback and responses, and inventory: so as to ensure transparent, and traceable transactions between the donors, staff, and admin. The development and testing of the web-based application demonstrates a platform built using modern technology including HTML, CSS, JavaScript, PHP, and MySQL that assisted in curating an effective solution that drastically improves the efficiency, transparency, and accuracy of in-kind inventory management operations in local community donation centres.

The research to uncover the truths about management of in-kind donations revealed the insufficient, unsecure, and error prone methods that are continuously used by multiple non-profit organisations, and this led to the development of a system that ensures low waste by setting expiry dates for perishable items in inventory and arranging it in the order of goods soon to expire. Moreover, the inadequacies in manual systems highlighted the need for an agile, adaptable system that aligns with the operations, aims and goals of a donation centre. The developed system provides real-time inventory visibility, ensures transparency for the donations' entire lifecycle, fosters effective communication between the donors and staff or admin, and provides a quick response to any actions performed resulting in the reduction of inventory loss, operational time, and errors in the operations of a local community donation centre. Finally, one of the biggest challenges the system tackles is the lack of integration between multiple processes in the operations of a donation centre, which causes delays in decision making and further fuels misallocation of resources, ultimately damaging their relationship between the donors and the centre, as well as hindering the operational efficiency of the centre.

The system developed integrates multiple processes through the various modules for each of the users, with the donor module handling donation inputs, assists donors in tracking their donations, and allows them to communicate their concerns to the other users. The staff module allows staff to process the donations, manage the inventory and distribute items, as well as communicate with the donors. Finally, the admin module manages all the users, ensuring only the right users gain

access to foster a work environment that will thrive under the changing pressures in the donation centres, and it allows them to communicate with donors as well as view inventory.

The system applies results from the research of various sources and systems developed by implementing arguments made, including the agreement from various sources on digitally transforming the non-profit sector from manual, paper-based methods that are error prone and hinder the data quality to a more digital and fully integrated solution which incorporates real-time tracking of inventory, documentation and categorisation of items, and the provision of dedicated donor portals that address the gap of disengagement in existing solutions. However, there is some contrast between the existing solutions reviewed and the developed system. While HELM and PlanStreet utilises high end technology and resources including Microsoft Business Intelligence Analytics, Unit4 for enterprise resource planning, and AI and predictive analytics, such tools are too complex for the project, which seeks to provide a system does not need extensive training, and doesn't incur additional operational costs for smaller non-profit organisations. Moreover, WASP's solutions requires purchasing proprietary hardware, which doesn't align with our goals of reducing the burden for centres, so the developed system tackles this by utilising the devices at hand, for instance, scanning QR code for distribution can be done with a simple mobile phone with an inbuilt camera that can scan the QR code. Finally, HELM prioritises on analytics and tactical decision making when managing in-kind donation inventory, neglecting communication with their respective donors, so this gap is recognised and addressed in the developed system.

The Agile-Scrum methodology was utilised in the development of the system, focusing on incremental and iterative development cycles, where there were sprints in the development and testing, ensuring each deliverable was up to the standards set by the requirements and system functionalities. Moreover, the system adheres to the Object Oriented Analysis and Design Approach by developing the system around entities that interact and execute their actions accordingly; however, multiple automated tests were omitted, and most were done manually to ensure that the system is fully functional, which slightly diminishes the robustness of the Scrum methodology.

The project successfully achieved its core functional requirements: the web-based in-kind inventory management system incorporates all the role-based functionalities such as user management for administrators, comprehensive inventory and donation management for staff, and

donation submission and tracking for donors. However, some of the non-functional requirements, including capacity, scalability, and reliability, weren't validated according to the plan stated in the methodology testing. This was as a result of technical skills gaps, however, to resolve this manual testing was done to validate majority of the non-functional requirements, which couldn't provide an objective data on whether the system met the capacity, scalability, or reliability requirements adequately, leaving the critical quality assurance specifications unmet. As a result, it left some of its objectives unanswered, more specifically, the system's ability to measure its success through the various testing methods that were omitted.

To conclude, even though the project's development and testing has some faults, the overall project delivers a meaningful contribution to society by providing a low-cost, accessible, and easy-to-use solution that replaces the error-prone and manual systems that are still utilised in local community donation centres. It offers unique and varying features, that connect the users, simplify and speed up operations, and ensures that the centre can meet their goals, which is to manage in-kind donations inventory effectively and efficiently.

6.2 Recommendations

The following recommendations are for future investigation and development, which explore and aim to solve the remaining operational and technical problems, including its accountability and transparency, its logical scalability, as well as the system's resilience; essentially addressing the objectives that weren't met as well as other areas of improvement.

6.2.1 System Testing

The reliance on manual testing left some critical non-functional requirements unaddressed, and partially validated. This leaves the system's ability to operate under high pressure uncertain. As a result, implementation and execution of automated testing is necessary to perform accurate and reliable stress and unit testing on the system's modules.

6.2.2 Further Inventory Categorisation and Enhanced Distribution

The current inventory tracking is rigid, linking the distributed items back to the donors' original submission with no full breakdown on the changes made and why. Moreover, staff are unable to distribute partial quantities, or combine quantities in the stock for one distribution, hindering the operational efficiency of the donation centres. As a result, the inventory structure needs to allow

item-level distribution, enable the staff to utilise partial donations, as well as include logistical categories such as weight, and size to aid in storage planning and management.

6.2.3 Live Chat Communication

While the feedback module is effective to raise concerns and pointers for improvement, the mode of communication is unsynchronised, leading to potential delays and unanswered questions. As a result, integrating a live chat feature for instant communication can foster a stronger connection and improve successful communication between the donors and the donation centre.

6.2.4 Incorporations of Multiple Centres into One System

Finally, the system is built as a single solution to just one centre, failing to solve the issue of large scale resource allocation across multiple regions of a country. As a result, developing an interconnect system that is capable of connecting donors to numerous centres will increase efficiency and can impact more multiple centres, maximising resource allocation efficiently on a wider scale.

References

(n.d.).

Akkerman, R., Buisman, M., Cruijssen, F., Leeuw, S. d., & Haijema, R. (2023). Dealing with donations: Supply chain management challenges for food banks. *International Journal of Production Economics* , 1-12.

Azman, M. A. (2024). *Inventory Management System.* Ipoh: Politeknik Ungku Omar.

HELM - Humanitarian Software. (2025). From Humanitarian Software: https://www.humanitariansoftware.com/

Holloway, S. (2025, February 12). *Inventory Management as a Key Driver of Sustainability in Supply Chains.* From Social Science Research Network (SSRN): https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5135338

Inventory Management Software. (2025). From PlanStreet: https://www.planstreet.com/inventory-management

Irere, I. (2023, October 18). *Adoption Of Modern Technology On Inventory Management Efficient Of Distributted Company Of Wasoko Ltd, Kigali, Rwanda.* From SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5199969

Islam, M. M. (2013). *In-Kind Donation Practice, Challenges and Strategies for NGOS and Donors.* Atlanta: Georgia Institute of Technology.

Jayadi, E. L. (2025). The digitalization of the humanitarian supply chain performance management literature and practice. *Journal of Humanitarian Logistics and Supply Chain Management*, 3-4.

Kaberi, J. M. (2013). *Organizational Systems and Programs Performance in Kenya Non-Governmental Organizations.* Nairobi: University of Nairobi.

Kesavan, R., Gay, D., Thevessen, D., Shah, J., & Mohan, C. (2023). *Firestore: The NoSQL Serverless Database for the Application Developer.* Institute of Electrical and Electronics Engineers (IEEE).

Lin, C.-Y., & Hsiung, H.-H. (2024). Enhancing donor engagement: Assessing the impact of online donation convenience on the willingness to donate to non-profit organizations. *Journal of Infrastructure, Policy and Development*, 1-2.

M.R., K., M.K., H., A., S., R., S., Y., K., & A., M. (2013). Object-oriented Modelling for Module-based Production Logistics Inventory System. *Australian Journal of Basic and Applied Sciences*, 555-562.

Muniandy, T. A., & Arbaiy, N. (2022, November 30). Development of Information Management System for Food Donation Distribution. *Applied Information Technology And Computer Science*, pp. 1-3.

Ndei, F. M., & Mutuku, M. (2021). Electronic Procurement and Performance of Non-Governmental Organizations in Kenya. Empirical Evidence from Pathfinder International, Kenya. *Journal of Business and Management Sciences*, 1-9.

Non Profit Inventory Management Made Simple. (2025). From Wasp Barcode Technologies: https://www.waspbarcode.com/inventory-control/non-profit-inventory-management

Penta, H. (2004). *A Comprehensive Testing Approach Using Jest For React Native Mobile Applications.* Northridge: California State University.

Sachdeva, S. (2016). Scrum Methodology. *International Journal Of Engineering And Computer Science*, 1-8.

Saïah, F., Vega, D., & Kovács, G. (2023). Toward a common humanitarian supply chain process model: the Frontline Humanitarian Logistics Initiative. *International Journal of Operations & Production Management*, 1-12.

Sanmocte, E. M., & Costales, J. A. (2025). Exploring Effectiveness in Software Development: A Comparative Review of System Analysis and Design Methodologies. *International Journal of Computer Theory and Engineering, Vol. 17, No. 1*, 36-43.

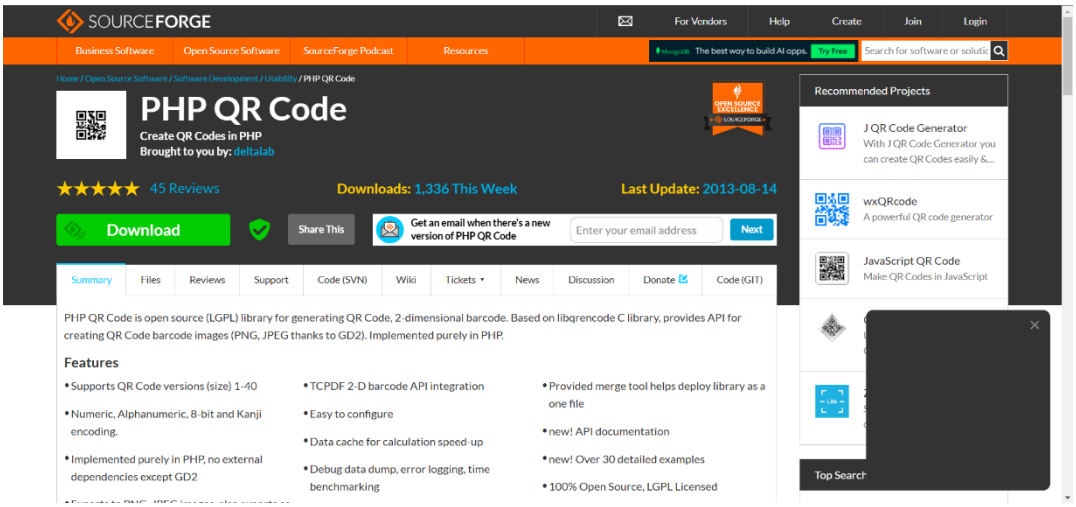
Shapiro, S. J., Oystrick, V., Liu, A. (., & Vadlamudi, N. K. (2024). From Spreadsheets to Scalability: Databases as a Next Step for Non-Profits. *Canadian Journal of Program Evaluation*, 1-8.

Sigala, I. F., Kettinger, W. J., & Wakolbinger, T. (2020). Digitizing the field: designing ERP systems for Triple-A humanitarian supply chains. *Journal of Humanitarian Logistics and Supply Chain Management*, 1-5.

Yadav, C., Dhakad, R., ., A., Panchal, M., & Kaur, E. B. (2024, August 28). *Revolutionizing Near-by Accommodation: An in-depth Analysis of React.js, Node.js, MongoDB, and Express.js Integration for Website Development.* From SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4932790

Appendices

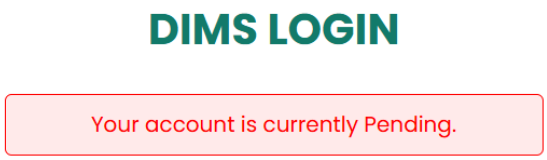
Appendix i: PHP QR Code



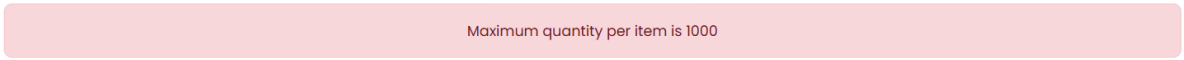
(PHP QR Code download | SourceForge, 2025)

Error Message Appendices

Appendix 1: Staff/admin login error

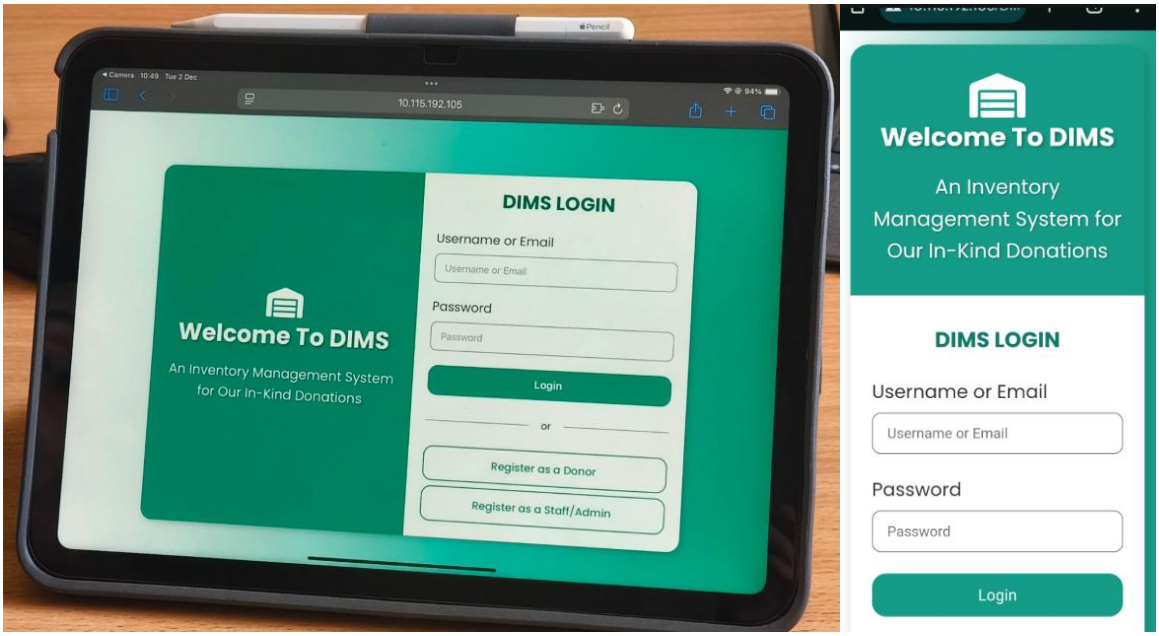


Appendix 2: More than max quantity error



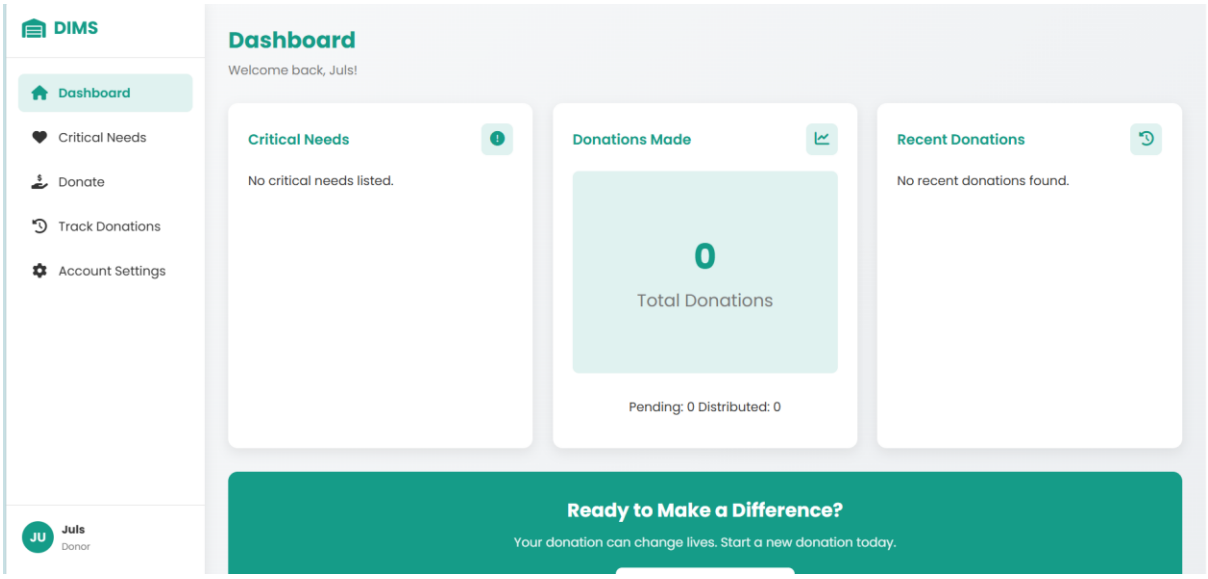
Non-Functional Requirements Appendices

Appendix 3: Portability



Test Case Result Appendices

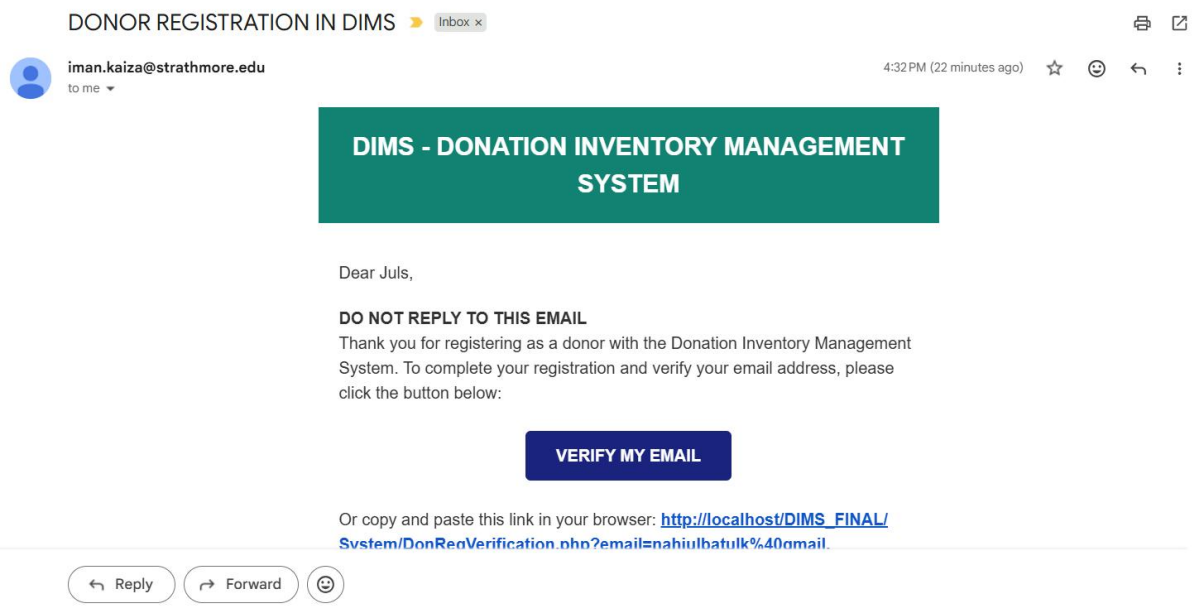
Appendix A: Successful Donor Login



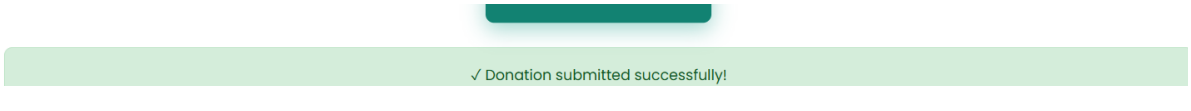
Appendix B: Registration Reflected in Database

	user_id	role	firstName	lastName	email	username	password_hash	contactDetails
Edit Copy Delete	1	Donor	*NULL*	*NULL*	nahjulbatulk@gmail.com	Juls	$2y$10$GQUE.Sb/wxm.Ka/sF6Ix7.fmDtasT7.C3Al2T4MXgId...	0706055671

Appendix C: Successful Email Verification Sent

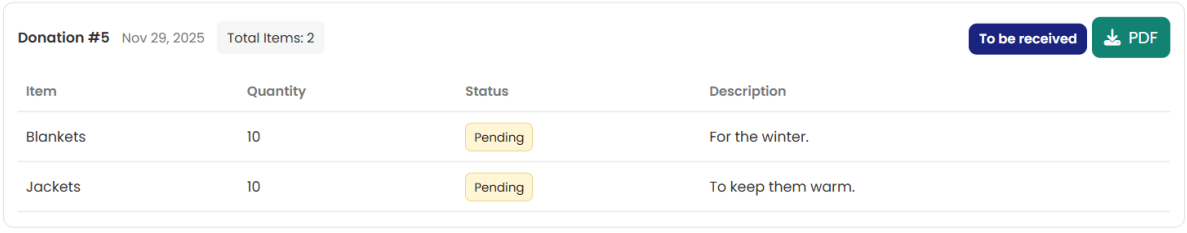


Appendix D: Donation Form

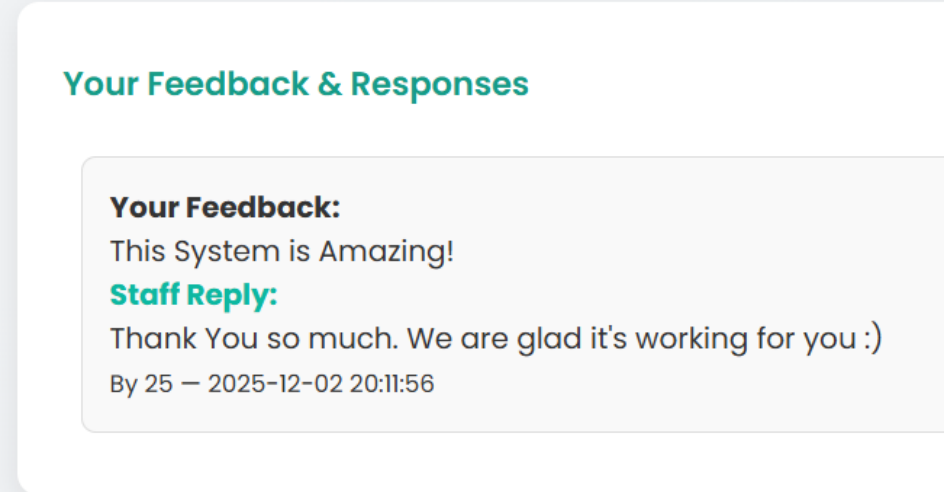


	item_id	donation_id	item_name	quantity	description	expiry_date	itemStatus
Edit Copy Delete	5	5	Blankets	10	For the winter.	*NULL*	Pending
Edit Copy Delete	6	5	Jackets	10	To keep them warm.	*NULL*	Pending

Appendix E: Donation Tracking



Appendix EA: Donor Feedback



Appendix EB: Donor Report Generation

All Donations Report

Generated: November 29, 2025, 4:36 pm

Donation #6 | Date: Nov 29, 2025 | Status: To be received

Total items: 1

Item Name	Quantity	Status	Description
Pads	10	Pending	Hygiene Products.

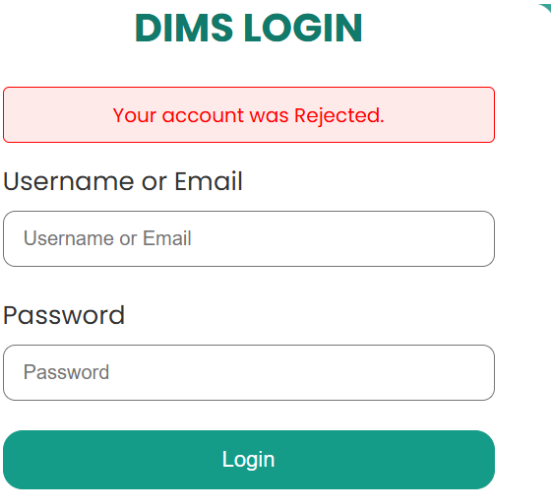
Donation #5 | Date: Nov 29, 2025 | Status: To be received

Total items: 2

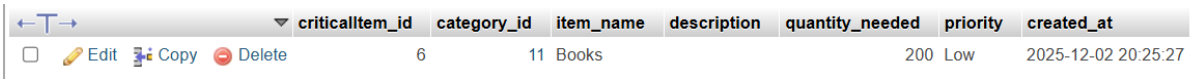
Item Name	Quantity	Status	Description
Blankets	10	Pending	For the winter.
Jackets	10	Pending	To keep them warm.

Appendix F: Successful Staff Login

Appendix G: Rejected Staff Login

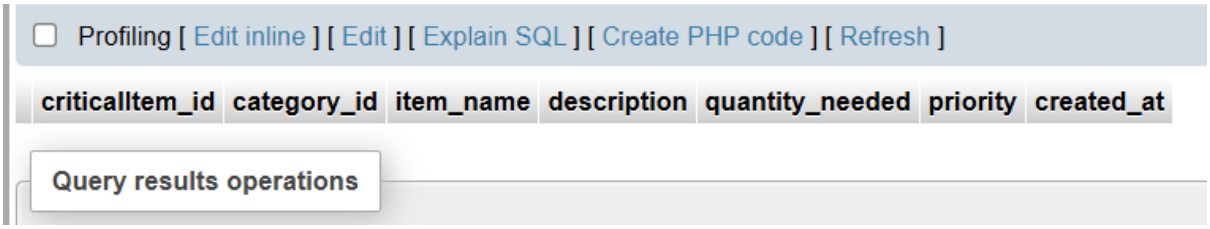


Appendix H: Adding a Critical Item

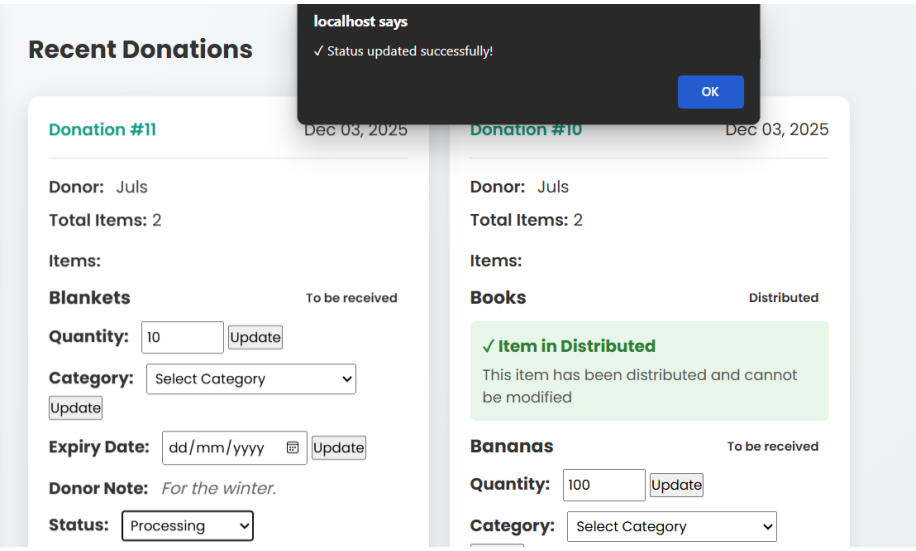


	criticalItem_id	category_id	item_name	description	quantity_needed	priority	created_at
Edit Copy Delete	6	11	Books		200	Low	2025-12-02 20:25:27

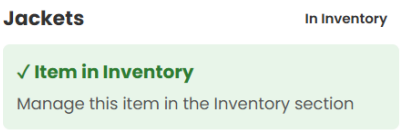
Appendix I: Deleting a Critical Item

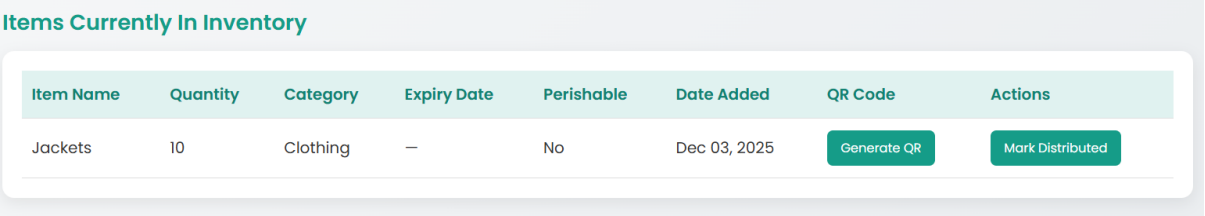


Appendix J: Editing Donation Status

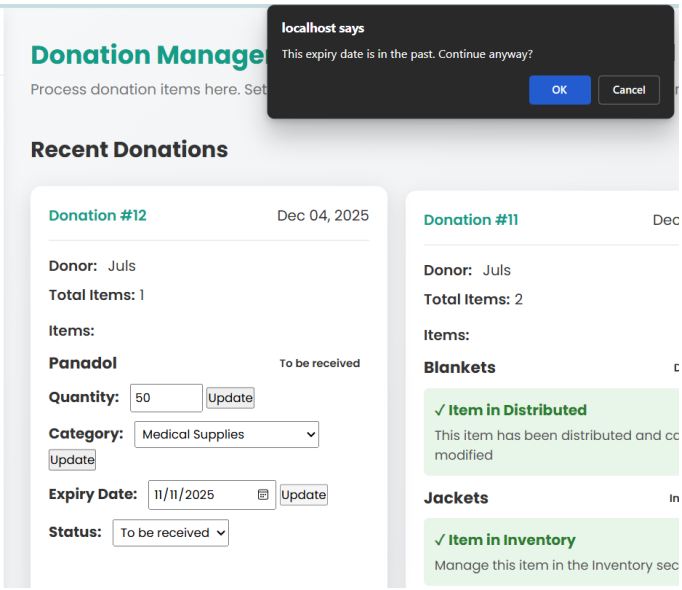


Appendix K: When Donation Status is 'In Inventory'

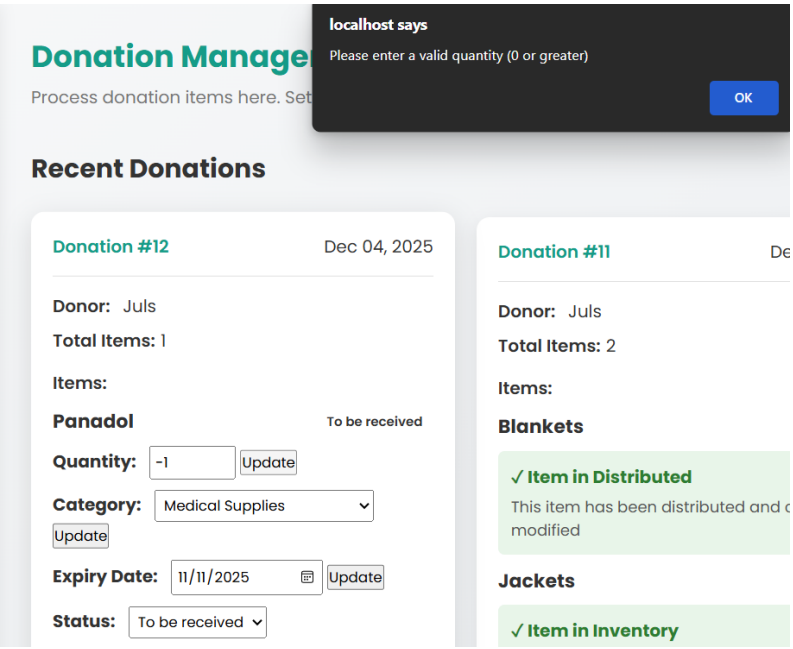




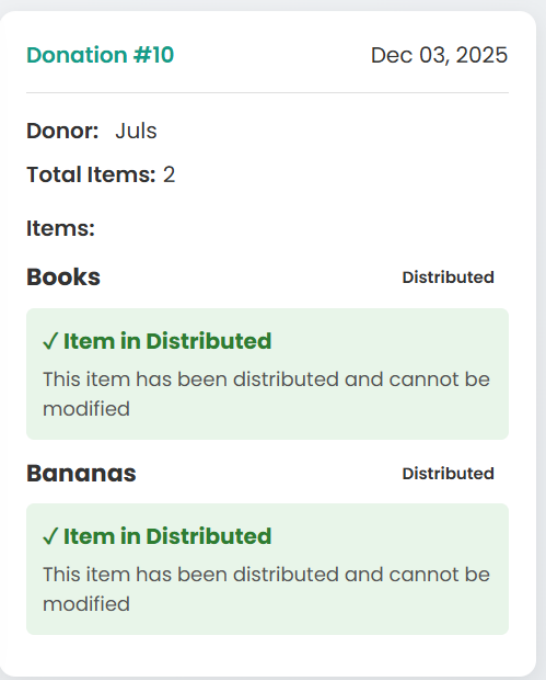
Appendix KA: Past Date as an Expiry Date



Appendix KB: Negative Quantity Error



Appendix KC: Cannot Change Donation Status Once Distributed

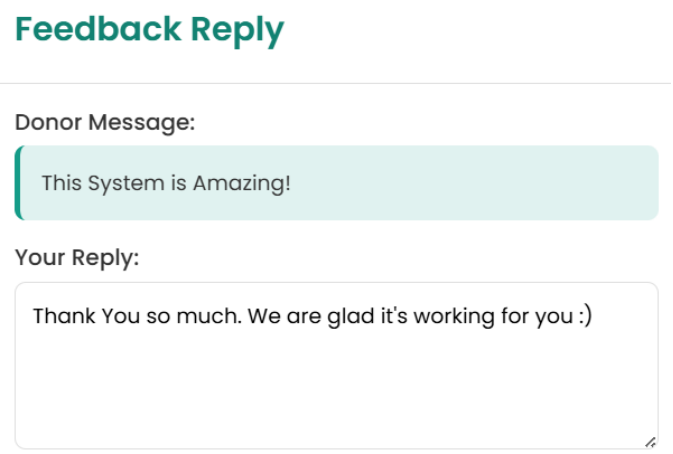


Appendix L: Staff Report Generation

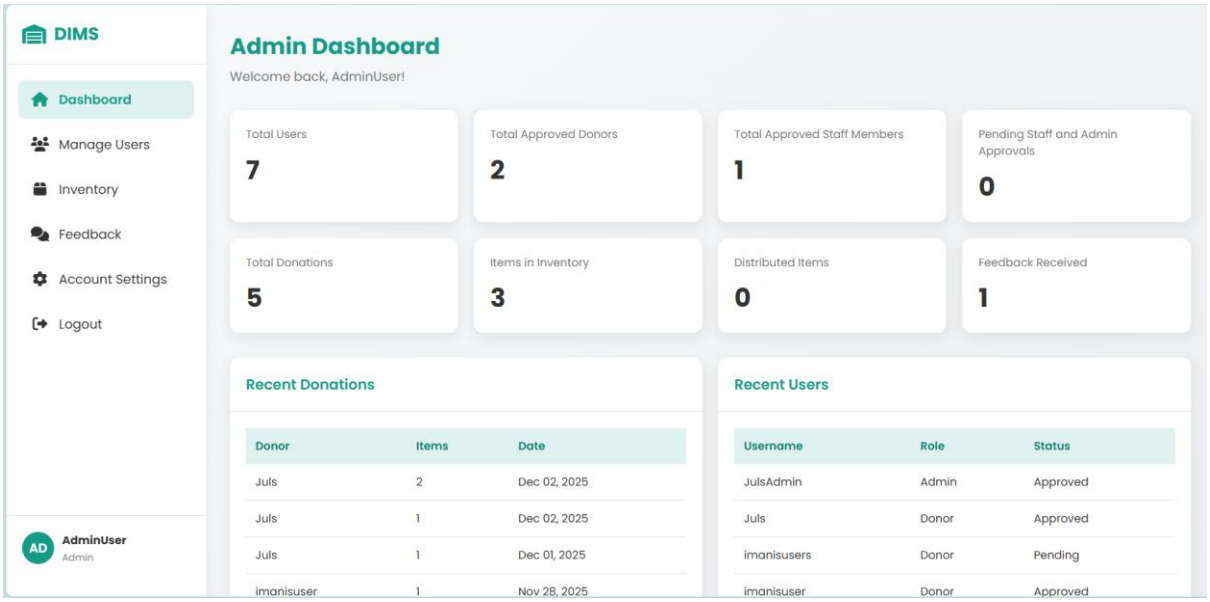
Full Inventory Report

Item Name	Quantity	Category	Expiry	Added	Distributed	Status
Blankets	10	Textiles	-	2025-12-03 23:47	2025-12-03 23:48	Distributed
Bananas	100	Food	2025-12-10	2025-12-03 23:46	-	In Inventory
Jackets	10	Clothing	-	2025-12-03 23:30	-	In Inventory
Books	50	Office/Stationary Su	-	2025-12-03 19:55	2025-12-03 19:57	Distributed

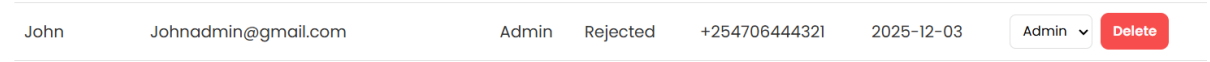
Appendix M: Staff Feedback

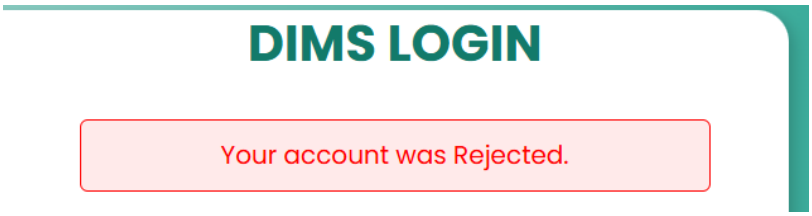


Appendix N: Successful Admin Login

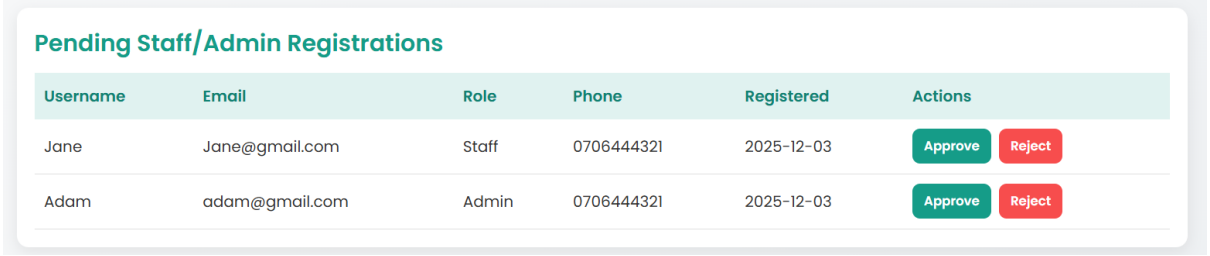


Appendix NA: Rejected Admin User

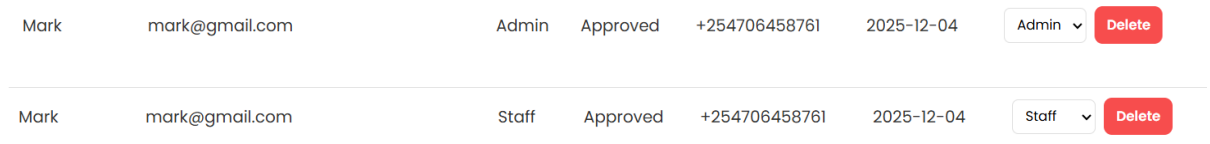




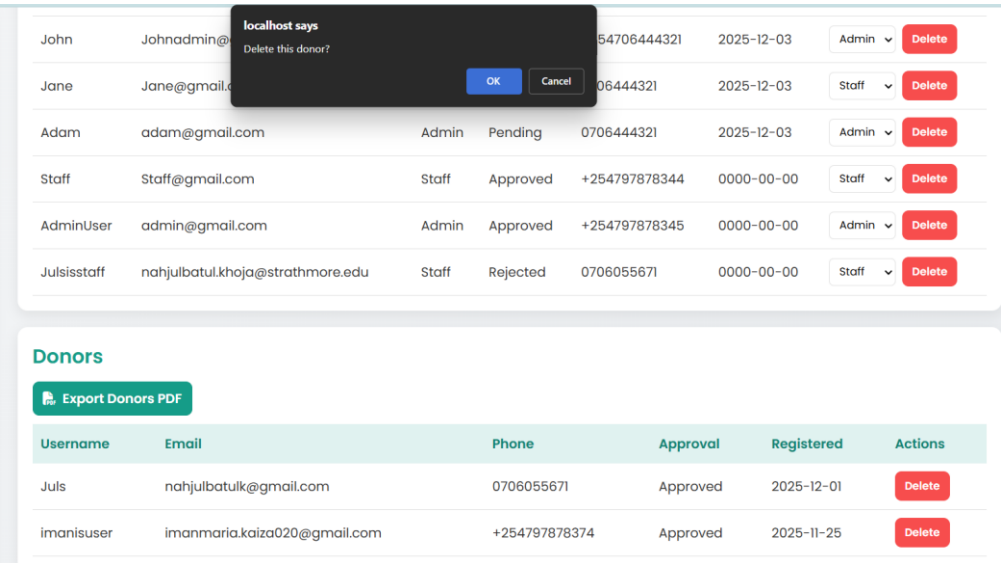
Appendix O: User Approvals

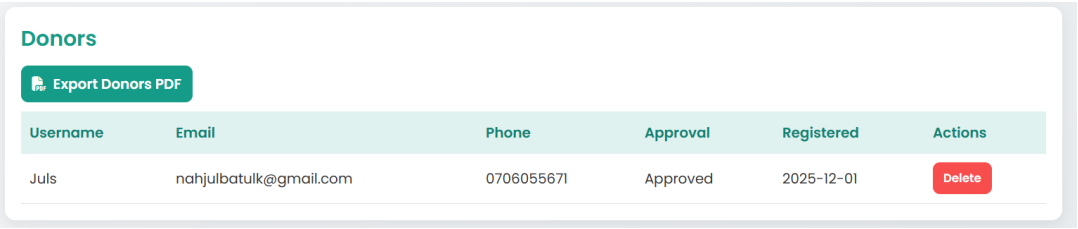


Appendix P: Role Changing

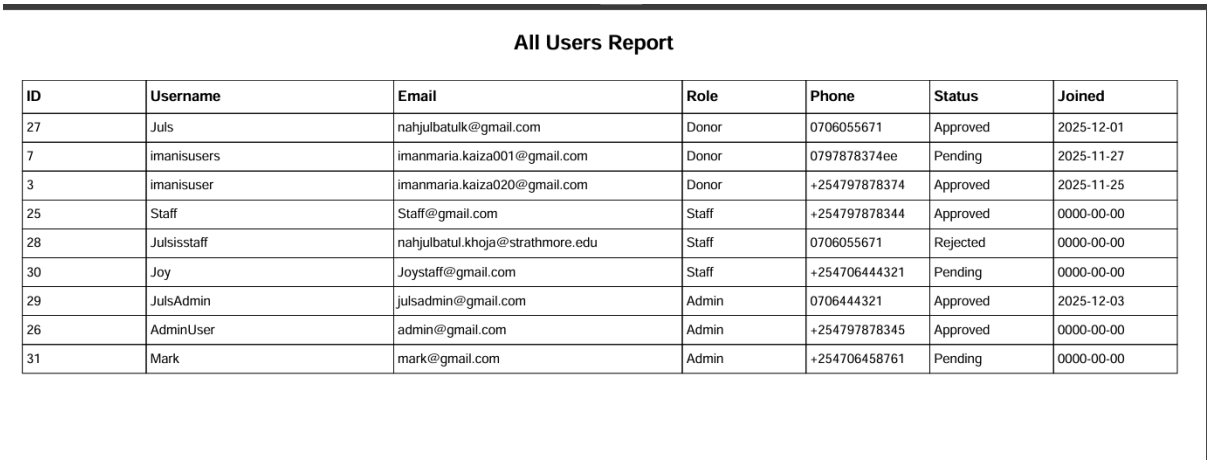


Appendix Q: Deleting a Donor User





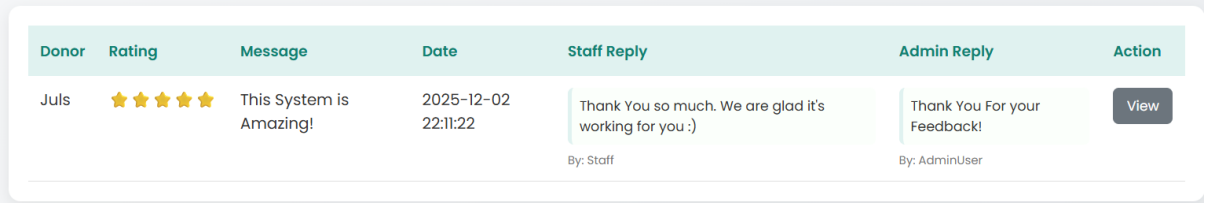
Appendix R: Admin Report Generation



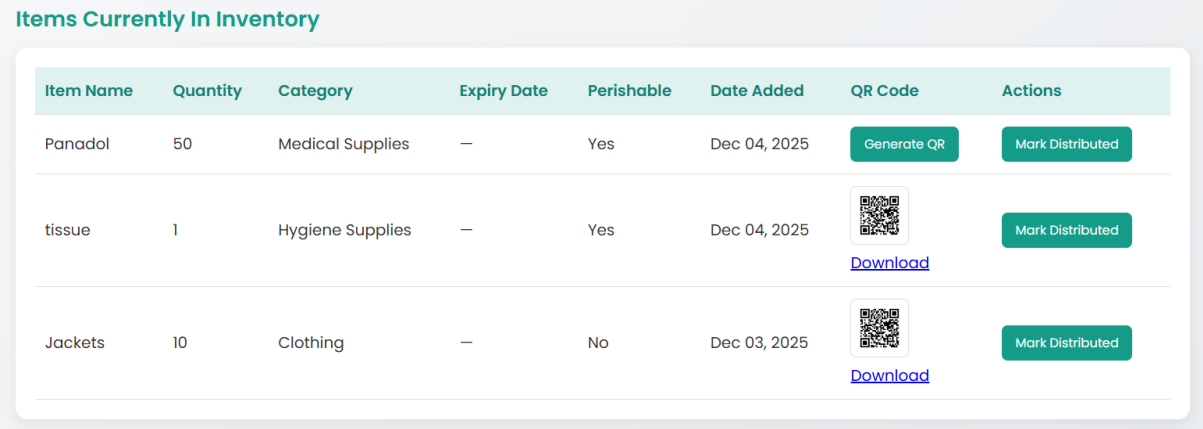
All Users Report

ID	Username	Email	Role	Phone	Status	Joined
27	Juls	nahjulbatulk@gmail.com	Donor	0706055671	Approved	2025-12-01
7	imanisusers	imanmaria.kaiza001@gmail.com	Donor	0797878374ee	Pending	2025-11-27
3	imanisuser	imanmaria.kaiza020@gmail.com	Donor	+254797878374	Approved	2025-11-25
25	Staff	Staff@gmail.com	Staff	+254797878344	Approved	0000-00-00
28	Julsisstaff	nahjulbatul.khoja@strathmore.edu	Staff	0706055671	Rejected	0000-00-00
30	Joy	Joystaff@gmail.com	Staff	+254706444321	Pending	0000-00-00
29	JulsAdmin	julsadmin@gmail.com	Admin	0706444321	Approved	2025-12-03
26	AdminUser	admin@gmail.com	Admin	+254797878345	Approved	0000-00-00
31	Mark	mark@gmail.com	Admin	+254706458761	Pending	0000-00-00

Appendix S: Admin Feedback



Appendix T: System QR Code Generation



Appendix U: System QR Code Scanning

